JOC The Journal of Organic Chemistry

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Total Synthesis, Structure Revision, and Neuroprotective Effect of Hericenones C–H and Their Derivatives

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Cite This: J. Org. Chem. 2021, 86, 2602–2620 Read Online

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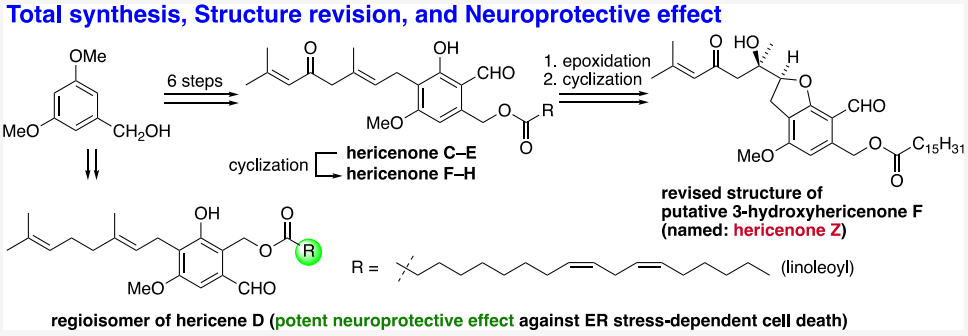

ABSTRACT: The first total syntheses of hericenones C–H and "putative 3-hydroxyhericenone F" were achieved. Highlights of the synthesis include the straightforward construction of the resorcinol core and geranyl side chain, assembly of the natural product skeleton by sequential O-geranylation and a clay/zeolite-mediated O → C rearrangement reaction, and a biomimetic cyclization to form a variety of bicyclic natural hericenones and their congeners. The structure of the "putative 3-hydroxyhericenone F" was revised as the 5-*exo* cyclization product (named: hericenone Z) of epoxyhericenone C through in-depth analyses of the cyclization modes in addition to NMR spectroscopic studies. To gain insights into the biological functions of geranyl-resorcinols in *Hericium erinaceus*, potential neuroprotective effects against endoplasmic reticulum (ER) stress-dependent cell death were evaluated systematically to clarify a fundamental structure–activity relationship. Among the compounds assayed, the linoleate-containing hericenone analogue, i.e., the regioisomer of hericene D, was found to possess the most potent neuroprotective effect against tunicamycin and thapsigargin-induced ER stress-dependent cell death.

■ INTRODUCTION

Hericium erinaceus (*H. erinaceus*) is known historically as a medicinal mushroom with a variety of nutritional and health-promoting effects.[1−3] By consuming the mushroom fruiting bodies or extracts, numerous positive physiological effects have been reported on the nervous, digestive, circulatory, and immune systems.[4] In particular, nervous system effects[5,6] such as the stimulation of the nerve growth factor (NGF) gene expression and enhancement of the neurite outgrowth[7] are considered to be beneficial for the prevention and treatment of dementia,[8,9] a cognitive disorder occurring in super-aged societies in developed countries. Some of the clinical testing data show that the continual oral intake of tablets containing dry samples of *H. erinaceus* led to improved cognitive functions.[10] Although certain kinds of mushrooms are known to exhibit beneficial medicinal effects for humans, such a cognitive improvement effect is highly characteristic to *H. erinaceus*. In this regard, many researchers have searched for specific active ingredients in this mushroom.[1,2,11−14]

Geranyl-resorcinols are representative low-molecular-weight chemical components characteristically included in the fruiting bodies of *H. erinaceus*. More than 30 congeners have been isolated so far and their biological activities are extensively investigated.[1,2] Due to their exceptional structural diversity and broad biological activities including antitumor,[15,16] inhibition of platelet aggregation,[17] antioxidant activity,[18] α-glucosidase inhibition,[19,20] neuritogenesis,[21] and neuroprotection,[22] we elected to pursue the collective total synthesis of hericenones C–H and explore the structure–activity relationships (SAR) of their derivatives.[23−26]

Structurally, the geranyl-resorcinols isolated from *H. erinaceus* can be classified into four groups, from Type 1 to Type 4, on the basis of an oxidation level of a geranyl side chain, in addition to the involvement of the fatty acid chain

Received: November 10, 2020
Published: January 25, 2021

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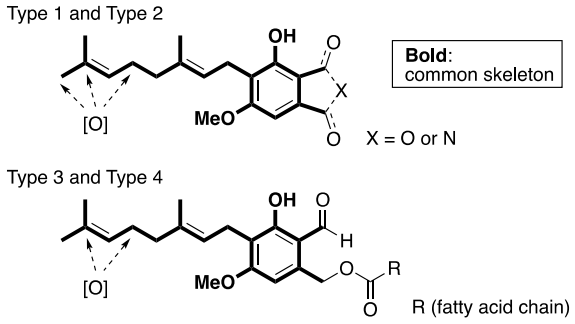
https://dx.doi.org/10.1021/acs.joc.0c02681
J. Org. Chem. 2021, 86, 2602–2620

(Figure 1). In our previous studies, we have synthesized more than 15 naturally occurring geranyl-resorcinols that were

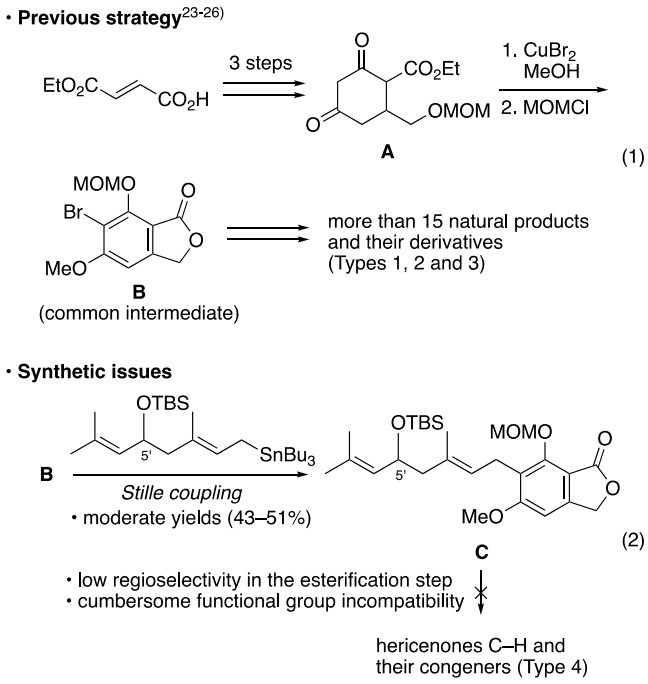


	oxygen[a]	fatty chain	examples
Type 1	no	no	hericenone J, isohericerin, hericenol A
Type 2	yes	no	hericenone A, B, I, erinacerin A–L
Type 3	no	yes	hericene A–D
Type 4	**yes**	**yes**	**hericenone C–H, 3-hydroxyhericenone F**

Figure 1. Classification of geranyl-resorcinols based on the chemical structure.

categorized into Types 1−3 with phthalide **B** as the common synthetic intermediate (Scheme 1, eq 1).[23−27] The compound

Scheme 1. Previous Studies



B was prepared by a $CuBr_2$-mediated one-pot multi-step reaction from the diketone **A**.[23] However, the natural products classified into Type 4 such as hericenones C−H and 3-hydroxyhericenone F could not be synthesized from the intermediate **B** due to three main issues: (1) the unsatisfactory Stille coupling between the double ortho-substituted aryl bromide **B** and C5′ silyloxy-containing side chain,[25] (2) the low regioselectivity in the esterification step,[23] and (3) the cumbersome functional group incompatibility at later stages of the synthesis (Scheme 1, eq 2). Therefore, an alternative synthetic strategy to obtain the natural products categorized in "Type 4" were developed as described herein.

Figure 2 summarizes the targets of this study. Hericenones C−E (**1a**, **1b**, and **1d**) are the original molecules reported exhibiting NGF-expression stimulating activity.[28] Sabaratnam et al. reported that hericenone E (**1d**) was most active among these hericenones, which potentiated NGF-induced neuritogenesis in PC12 cells.[21] Their cyclized derivatives, i.e., hericenones F−H (**2a**, **2b**, and **2d**), were also discovered from the fruiting bodies of *H. erinaceus* and hericenone H (**2d**) is reported to stimulate NGF-expression in astroglial cells more strongly than hericenones F (**2a**) and G (**2b**).[29] In addition, 3-hydroxyhericenone F (**2e**) exhibits neuroprotection against endoplasmic reticulum (ER) stress-dependent cell death.[22] These biological activities on the nervous system through the consumption of *H. erinaceus* make the hericenones attractive targets in drug discovery for neurodegenerative disorders such as Alzheimer's and Parkinson's diseases. In our previous study, the 5′-deoxo congeners, hericenes A−C (**3a**−**3c**) and their regioisomers (**4a**−**4c**) with respect to aldehyde and the fatty ester moiety, were synthesized to clarify the primary SAR on the neuroprotection against ER stress-dependent cell death.[26] It was found that the regioisomers of hericenes B (**4c**) and C (**4b**) showed neuroprotection against tunicamycin-induced ER stress at a concentration of 10 μM,[26] which was comparable to the activity of **2e**.[22] The SAR study also suggested that the structure of the fatty acid components slightly affected the neuroprotective effect; the unsaturated ester, i.e., oleates (**3c** and **4c**), tended to be more active than the saturated esters (**3a**, **3b**, **4a**, and **4b**). This observation prompted us to evaluate the activity of the linoleate esters (**d** series). Herein, we report the neuroprotective effects of the linoleate-containing synthetic molecules (**d** series) in addition to the first total syntheses of hericenones C−H, 3-hydroxyhericenone F, and their congeners. Importantly, our findings have revealed that the original structure reported as 3-hydroxyhericenone F (**2e**) should be revised as structure **22a**, as formed by the 5-*exo* cyclization of epoxyhericenone C (**21a**).

■ RESULTS AND DISCUSSION

Considering the synthetic issues described above and difficulties to connect a side chain with a resorcinol core by the Stille coupling method (Scheme 1, eq 2), we decided to introduce a fatty ester moiety at the beginning of the synthesis and use O → C rearrangement,[27c,e,30−38] to insert the geranyl side chain at the ortho position of the phenolic hydroxyl group. Thus, our second-generation synthesis began with esterification of alcohol **5**, prepared by reduction of commercially available 3,5-dimethoxybenzoic acid (Scheme 2).[39] The Vilsmeier−Haack reaction of the resulting fatty acid esters **6a**−**6d** provided aldehydes **7a**−**7d** in 83−98% yields with complete regiocontrol. The methyl ether ortho to the formyl group was selectively cleaved by BBr_3 to afford phenols **8a**−**8d** in 95−100% yields. After short screening of the solvent, MeCN was found optimal for deprotection with BBr_3. When conventional dichloromethane (DCM) was used for the unsaturated esters **7c** and **7d**, the addition of amylene was necessary to avoid hydrobromination (or bromination) of the fatty ester moiety caused by generating HBr and Br_2 (probably produced from BBr_3 and trace amounts of water). It is noteworthy that chromatographic purification was unnecessary over three steps for the crystalline saturated esters (**a** and **b** series). The resultant phenols **8a**−**8d** were O-alkylated with

https://dx.doi.org/10.1021/acs.joc.0c02681
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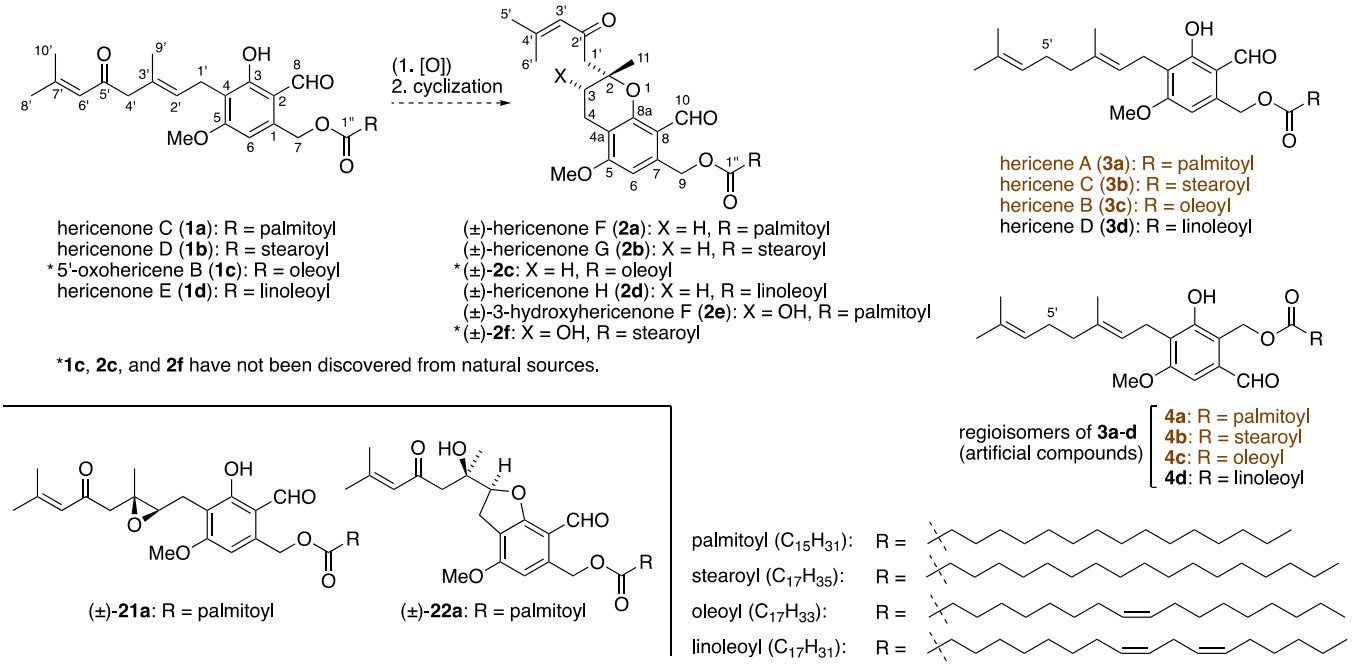
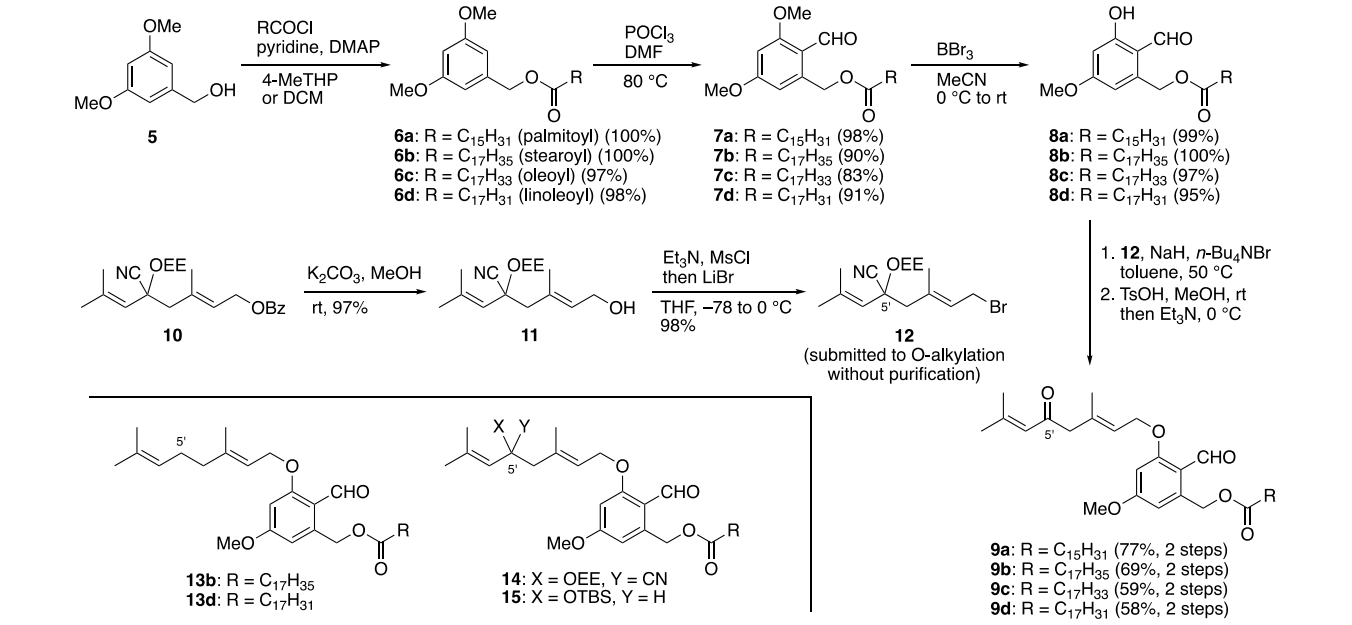


Figure 2. Structures of hericenones C–H and their congeners. Compounds synthesized in a previous study[26] are shown in brown letters.

Scheme 2. Syntheses of the Rearrangement Precursors 9a–9d



the side chain **12** prepared from benzoate **10**[25] in two steps under the treatment with NaH in the presence of n-Bu_4NBr as a phase transfer catalyst. Next, the acid-catalyzed removal of the ethoxyethyl (EE) group followed by elimination of cyanide with Et_3N afforded the rearrangement precursors **9a–9d** in 58–77% yields for two steps. Regarding the assembly of the core and the side chain, direct C-geranylation[38,40] of **8a** with geranyl bromide as a model substrate was investigated earlier. However, the desired C-geranylated product (i.e., **3a**) was detected in only 10% yield (confirmed by the crude NMR spectrum) even under the optimized conditions using excess LiH and catalytic n-Bu_4NBr with heating at 100 °C for 38 h (45% conversion, C-alkylation/O-alkylation = 22:78). Alternative dehydrative coupling between the resorcinol core and geraniol using $BF_3{\cdot}Et_2O$ or $Sc(OTf)_3$[41,42] failed to give the desired C-geranylation product.

With the precursors **9a–9d** in hand, the crucial O → C rearrangement was surveyed (Table 1). In prior to testing **9a–9d**, the readily available C5′-deoxo congeners **13b** and **13d** were subjected to rearrangement for optimization of conditions (entries 1–9). Literature methods using montmorillonite K10 (M-K10) clays[31,33,35–37] resulted in the formation of the degeranylated phenol **8b** in preference to the desired product **3b** (hericene C) (entry 1). Hence, extensive conditions were explored to improve the selectivity for **3**. Among the solid additives tested, Florisil[32] and molecular sieves 13X (MS13X)[43] turned out promising. However, in the case of Florisil, the product ratios varied considerably, probably

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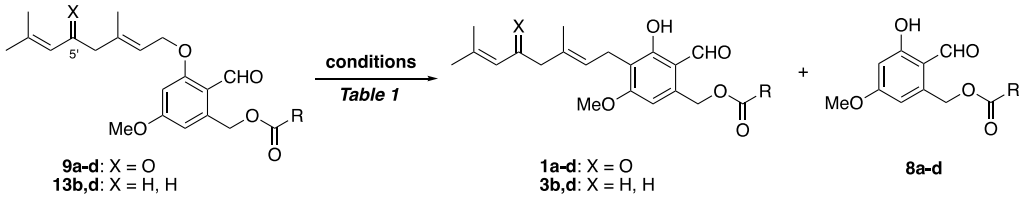
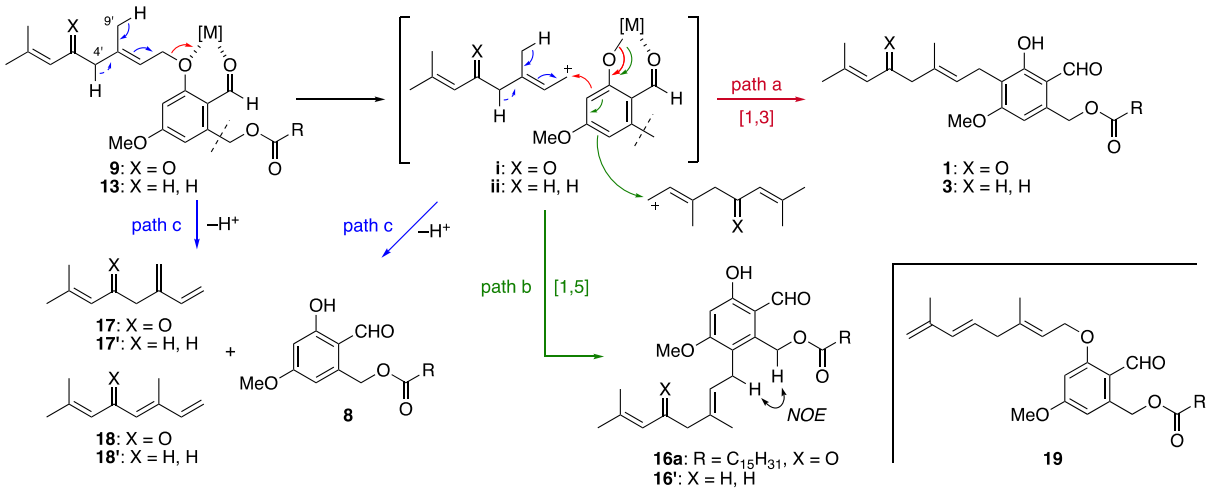
Table 1. O → C Rearrangement Promoted by Solid Additives[a]



entry	substrate	additive	solvent	temp (°C)	time (h)	content ratio (%)[b] 9 or 13	1 or 3	8	mass balance (%)[c]
1	**13b**	M-K10 (1 wt equiv)	toluene	0 to rt	17	13	24	63	84
2	**13b**	Amberlyst 15(H) (1 wt equiv)	toluene	rt	2.5			100	95
3	**13d**	$FeCl_3$-alumina[d] (3 equiv)	toluene	−78	0.8	2	38	60	[e]
4	**13b**	silica gel 40−50 μm (10 wt equiv)	toluene	110	21		33	67	19
5	**13b**	Florisil 150−250 μm (10 wt equiv)	toluene	110	3−16		0−55	45−100	64−84
6	**13d**	MS13X (pellets) (3 wt equiv)	toluene	100	14		40	60	96
7	**13d**	MS13X (powdered)[f] (3 wt equiv)	toluene	100	14		79	21	73
8	**13b**	MS13X (powdered)[f] (10 wt equiv)	toluene	100	16		67	33	27
9	**13d** (1.7 g)	MS13X (powdered)[f] (3 wt equiv)	toluene	100	5		80 (34)	20	65
10	**9a**	MS13X (powdered)[f] (4 wt equiv)	toluene	100	24	30	30[g]	40	68
11	**9a**	M-K10 (1 wt equiv)	toluene	rt to 50	4	6	20	74	85
12	**9a**	M-K10 (5 wt equiv)	toluene	rt	95	30	16	54	78
13	**9a**	M-K10 (5 wt equiv), MS4A (powdered)[f] (10 wt equiv)	toluene	rt	47	28	30	42	68
14	**9a**	same as entry 13	toluene	50	24		26	74	100
15	**9a** (1.0 g)	same as entry 13	DCE	50	48		(33)	(36)	[h]
16	**9b** (0.5 g)	same as entry 13	DCE	50	48		(31)	(47)	[h]
17	**9c** (0.5 g)	same as entry 13	DCE	50	48		(30)	(41)	[h]
18	**9d** (0.5 g)	same as entry 13	DCE	50	48		(28)	(42)	[h]
19	**9a**	Amberlyst 15(H) (1 wt equiv)	toluene	0 to rt	48		20	80	65
20	**9a**	$FeCl_3$-alumina[d] (2 equiv)	toluene	0	5		19	81	99
21	**9a**	Florisil 150−250 μm (10 wt equiv)	toluene	110	24	6	12	82	71

[a]Each reaction was carried out with 50 mg of the starting material unless indicated otherwise. [b]Determined by ^{1}H NMR analysis after the removal of solid reagents by filtration. Values in parentheses refer to isolated yields after flash chromatography. [c]Weight percentage of the crude mixture relative to the starting material. [d]$FeCl_3$-alumina was prepared by mixing $FeCl_3$ (5 g) and basic alumina (25 g) in DCM (75 mL). [e]Many byproducts were detected on TLC. [f]Prepared by grinding commercially available molecular sieves (pellets). [g]A 2:1 mixture of **1a** (hericenone C) and **2a** (hericenone F). [h]Not determined.

Scheme 3. Plausible Mechanisms for O → C Rearrangement



due to the non-uniformity of chemical compositions in particle sizes (entry 5). In contrast, the rearrangement was reproducible in the case of MS13X. After many trials, it was found that at least 3 equiv (by weight) of MS13X were necessary for the conversion of all starting materials (entry 6). The rearrangement was enhanced when the well-ground, powdered MS13X were used, although the mass balance (the weight percentage for the crude mixture relative to the starting

https://dx.doi.org/10.1021/acs.joc.0c02681
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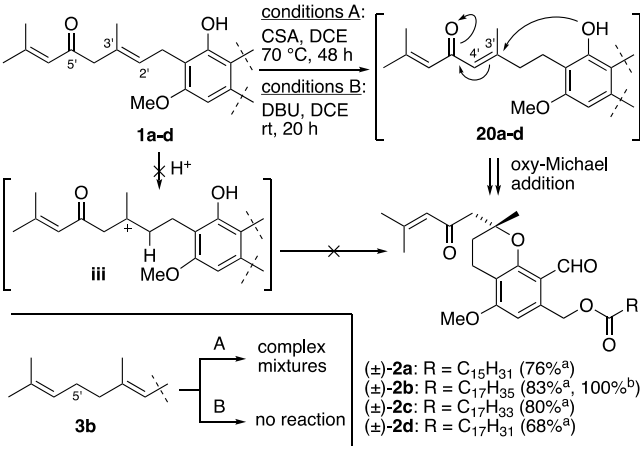
material) decreased (entry 7). The significant decrease of mass balance when using 10 equiv (by weight) of MS13X may arise from the trapping capability to capture the products in the interlamellar regions of zeolites (entry 8).[44] With optimal conditions identified by using 3 equiv of powdered MS13X, 1.7 g of **13d** was converted to **3d** (hericene D) in one batch with 34% isolated yield after chromatography (entry 9).

Next, the rearrangement was applied to **9a**−**9d** possessing the C5′-ketone (entries 10−21). Although the desired product **1a** (hericenone C) formed under the optimal conditions established for the C5′-deoxo series, cyclization to give hericenone F (**2a**) was partially accompanied by rearrangement (**1a**/**2a** = 2:1), along with the recovery of **9a** after heating for 24 h (entry 10). This result suggests that only a slight difference in the functional groups influences the efficiency of rearrangement. Therefore, optimization was again undertaken for **9a**, to promote rearrangement at lower temperature. After extensive trials, M-K10 was found superior to other activators. Although a proportion of **8a** was still high when using M-K10 clays alone (entries 11 and 12), a content ratio of **1a** slightly increased when 5 equiv (by weight) of M-K10 was used with molecular sieves 4 Å (MS4A) (entry 13). When the solvent was replaced by dichloroethane (DCE)[45] and the temperature was increased to 50 °C, a near 1:1 selectivity with respect to **1a** and **8a** was obtained (entry 15). It is important to note that the reactions could be scaled up to a 1 g level and 308 mg of **1a** was provided in one batch in 33% isolated yield after chromatography with the recovery of **8a** (36% yield).[46] The reaction was reproducible with little dependence on the fatty acid chain (entries 16−18). Although further improvements were not achieved, the recovered **8a**−**8d** could be recycled to increase the overall yields of **1a**−**1d**. Other conditions shown in entries 19−21 or the use of inorganic Lewis acid such as $FeCl_3$, $ZnCl_2$, Et_2AlCl, $InBr_3$, $Bi(OTf)_3$, $La(OTf)_3$, or $Sc(OTf)_3$ resulted in the predominant formation of **8a**−**8d** with none or trace amounts of **1** or **3**.

Scheme 3 shows plausible mechanisms for O → C rearrangement. It is likely that the phenolic oxygen is activated by metals in zeolites or clays to cleave the C−O bond after which C-alkylation takes place to provide the [1,3]-rearrangement products **1** or **3** (path a). Although a concerted [1,3]-shift mechanism is also possible,[47] isolation of the [1,5]-rearrangement product **16a** (2% yield) in the reaction of **9a** supports a cleavage−recombination process (path b). Formation of the phenol **8** is ascribed to the competed deprotonation from geranyl ether **9**/**13** or their allylic cation intermediates **i**/**ii** (path c). The elimination products **17** and **18** were isolated in 12 and 0.8% yields, respectively, in the reaction of **9a**.[48] To avoid elimination processes, by reducing the acidity of C4′−H, the C5′-keto-protected cyanohydrin ether **14** or the reduced TBS ether **15** was subjected to rearrangement (Scheme 2). However, only the corresponding C5′-ketone **9** resulted from **14**, and elimination of the TBS ether from **15** gave the triene **19** (Scheme 3), along with the cleavage of the C−O bond to give **8** in each case.

After achieving the straightforward total syntheses of hericenones C−E (**1a**, **1b**, and **1d**) and 5′-oxohericene B (**1c**) with minimal usage of protective groups, we embarked on the syntheses of hericenones F−H (**2a**, **2b**, and **2d**) and their congener (**2c**) by cyclization (Scheme 4). According to our previous work,[25] the synthetic hericenones **1a**−**1d** were treated with (+)-10-camphorsulfonic acid (CSA) in DCE (conditions A). Cyclization progressed gradually at 70 °C to give the

Scheme 4. Syntheses of Hericenones F−H (2a, 2b, 2d) and the Congener (2c)



[a]Isolated yield for conditions A. [b]Isolated yield for conditions B.

products (±)-**2a**−**2d** in 68−83% yields. It should be noted that cyclization occurred more smoothly at room temperature under basic conditions using DBU, affording (±)-**2b** quantitatively (conditions B). Interestingly, the 5′-deoxo analogue of **1b** (i.e., **3b**: hericene C) did not undergo cyclization under these conditions. These differences in reactivity led to the hypothesis that either concerted 6-*endo* cyclization or stepwise cyclization via the cationic intermediate **iii** was unlikely even under acidic conditions. The careful tracking of the reaction with **1b** revealed that isomerization of the double bond from C2′−C3′ to C3′−C4′ occurred (Table 2). The isomerization process was much faster under basic

Table 2. Cyclization of Hericenone D (1b)

				ratio (%)[a]		
entry	promotor	temp (°C)	time (h)	**1b**	**20b**	**2b**
1	CSA	rt	20	89	9	2
		→50	3	75	21	4
		50	16	32	27	41
2	DBU	0	1.5	60	9	31

[a]Determined by ^{1}H NMR after flash chromatography.

conditions than under acidic conditions (entry 1 vs 2). Given these experimental results in mind, a possible mechanism for cyclization was deduced (Scheme 4). Initially olefin isomerization occurs to give the conjugated dienone **20a**−**20d**, which undergoes intramolecular oxy-Michael addition to afford (±)-**2a**−**2d**. These experimental findings coincide with the fact that cyclization derivatives of hericenes A−D (**3a**−**3d**) lacking a ketone functionality have not been discovered from natural sources, while the ketone-containing bicyclic hericenones such as hericenones F−H (**2a**, **2b**, and **2d**), hericenone I,[22] and erinacerin A[49] have been identified upon isolation.

Next, our attention was focused on the synthesis of 3-hydroxyhericenone F [(±)-**2e**], a leading molecule reported to have neuroprotective activity. In theory, **2e** can be formed by the 6-*endo* cyclization of epoxyhericeneone C (**21a**), although it is anticipated that 5-*exo* cyclization would compete. Before investigating this cyclization, the needed regioselective epoxidation of the C2′−C3′ olefin was explored (Scheme 5). Although preliminary epoxidation of *o*-geranylphenol with *t*-

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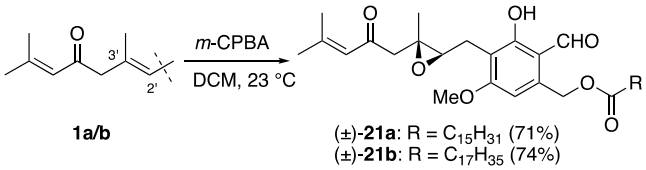
Scheme 5. Syntheses of Epoxyhericenones C [(±)-21a] and D [(±)-21b]

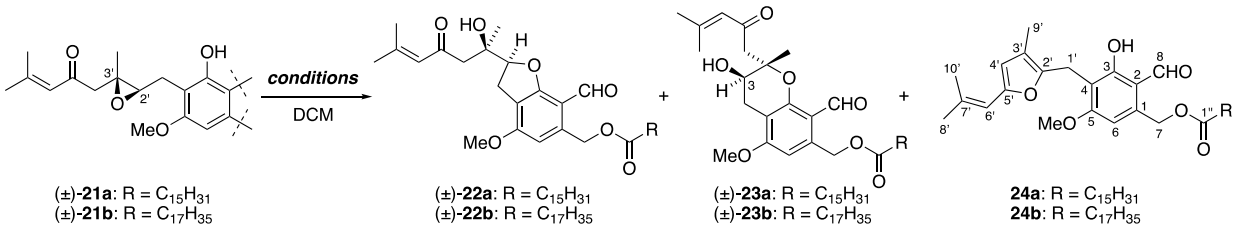


BuOOH and catalytic $VO(acac)_2$[50,51] proceeded in 85% yield (DCM, rt., 1 h), no reaction occurred when the conditions were applied to the fully functionalized **1a**. After several attempts, treatment of **1a/b** with anhydrous *m*-CPBA (3−4 equiv) was found to afford (±)-**21a/b** in 71−74% yields. As a minor product, the phenyl formate derivative resulted from Baeyer−Villiger oxidation of the aldehyde moiety (ca. 5% yield).[52]

With epoxides (±)-**21a/b** in hand, cyclization was attempted under acidic conditions with the aim that phenolic oxygen would attack the cationic C3′-carbon through an S_N1-like pathway (Table 3). When relatively strong sulfonic acids such as *p*-toluenesulfonic acid (TsOH) and CSA were used, contrary to our expectation, the major products obtained were not identical to the natural product (entries 1 and 2). After detailed analyses of the NMR spectra of products, and by considering the reaction modes and by derivatization to acetates (vide infra), one of the major products was identified as the 3-hydroxyhericenone F [(±)-**23a**] whereby the stereoconfiguration of C3 is opposite to the original assignment [(±)-**2e** vs (±)-**23a**]. Another major product identified was the furan derivative **24a** whose structure was fully determined by 2D NMR and mass spectroscopic methods. To determine why the "putative 3-hydroxyhericenone F (**2e**)" was not obtained, cyclization was studied in further detail. When the relatively mild trichloroacetic acid was used, a product showing the same spectra as the "putative 3-hydroxyhericenone F (**2e**)" was obtained in 31% yield (entry 3). Analysis of the NMR spectra suggested the structure to be consistent with the 5-*exo* cyclized product (±)-**22a**, thereby establishing the actual structure of "putative 3-hydroxyhericenone F (**2e**)" (vide infra). At this point, we noticed that, in some cases, 5-*exo* cyclization occurred to some extent during the epoxidation step. Therefore, thermal cyclization was tried in the absence of protic acid. Surprisingly, (±)-**22a** exclusively formed even though cyclization took more than 5 days to complete (entry 4). Although the real natural product (±)-**22a** was obtained with a maximum yield of 74%, this result could not be reproduced (25−74% yield). We suspected that the reproducibility was dependent on the purity of the epoxide (±)-**21**; a very small amount of benzoic acid derived from *m*-CPBA would affect the mode of cyclization. With this assumption, cyclization was carried out in the presence of benzoic acid. Remarkably, the 5-*exo* product (±)-**22b** formed reproducibly together with the furan byproduct **24b** (entry 5). Finally, cyclization was achieved under basic conditions on the idea that the phenoxide produced by deprotonation of (±)-**21** would attack the less hindered C2′-carbon through an S_N2-like pathway. As expected, the 5-*exo* products (±)-**22a/22b** were produced predominantly in 68−79% yields when treated with DBU, without formation of the furan byproducts **24a/24b** (entries 6 and 7). Careful monitoring of TLC suggested the formation of an unknown intermediate, which could be collected by flash chromatography but was immediately decomposed into furan **24a/24b** after dissolving it in $CDCl_3$ for NMR analysis (Figure S1 in the Supporting Information). This product was also detected by TLC during cyclization with benzoic acid, and its formation seemed concomitant to the formation of the furan (Figure S2 in the Supporting Information). Given these observations together, the unstable compound detected on TLC was probably a labile precursor to **24a/24b**.

Scheme 6 summarizes the mechanisms to produce **22**−**24**. Under strong acidic conditions using TsOH or CSA, S_N1-like cationic cyclization (formal 6-*endo* cyclization) to (±)-**23** predominated over 5-*exo* cyclization (path a). The relative stereochemistry for C2 and C3 in (±)-**23** can be deduced by the mode of cyclization in which the nucleophile attacks the more cationic C3′ (corresponding to C2 in **23**) from the back side of epoxide in conformation **v**. It should be noted that the relative stereochemistry proposed in this study is opposite to the original assignment (i.e., **2e**), thus giving a *cis* relationship for C3−OH and C2−Me.[53] The existence of a secondary alcohol was confirmed unambiguously by the observation of a doublet peak at δ 3.32, corresponding to C3−OH in the ^1H

Table 3. Cyclization of Epoxyhericenones C [(±)-21a] and D [(±)-22b]



entry	epoxide	promotor (equiv)	temp (°C)	time (h)	(±)-22a/22b (%)[a]	(±)-23a/22b (%)[a]	24a/24b (%)[a]
1	(±)-**21a**	TsOH·H_2O (1)	rt	0.5	<1[b]	36[b]	64[b]
2	(±)-**21a**	CSA (2.5)	0	1.5	<3	34	54
3	(±)-**21a**	CCl_3CO_2H (0.2)	50	5	31	17	52
4	(±)-**21a**		70	120	<74[c]		
5	(±)-**21b**	$PhCO_2H$ (6)	rt → 70	70	49		22
6	(±)-**21a**	DBU (4)	rt	2.5	68	4	3[d]
7	(±)-**21b**	DBU (4)	0	2	79	4	7[d]

[a]Isolated yield. [b]Determined by ^1H NMR analysis of crude mixture. [c]Not reproducible. [d]Detected after dissolving the isolated product in $CDCl_3$.

https://dx.doi.org/10.1021/acs.joc.0c02681
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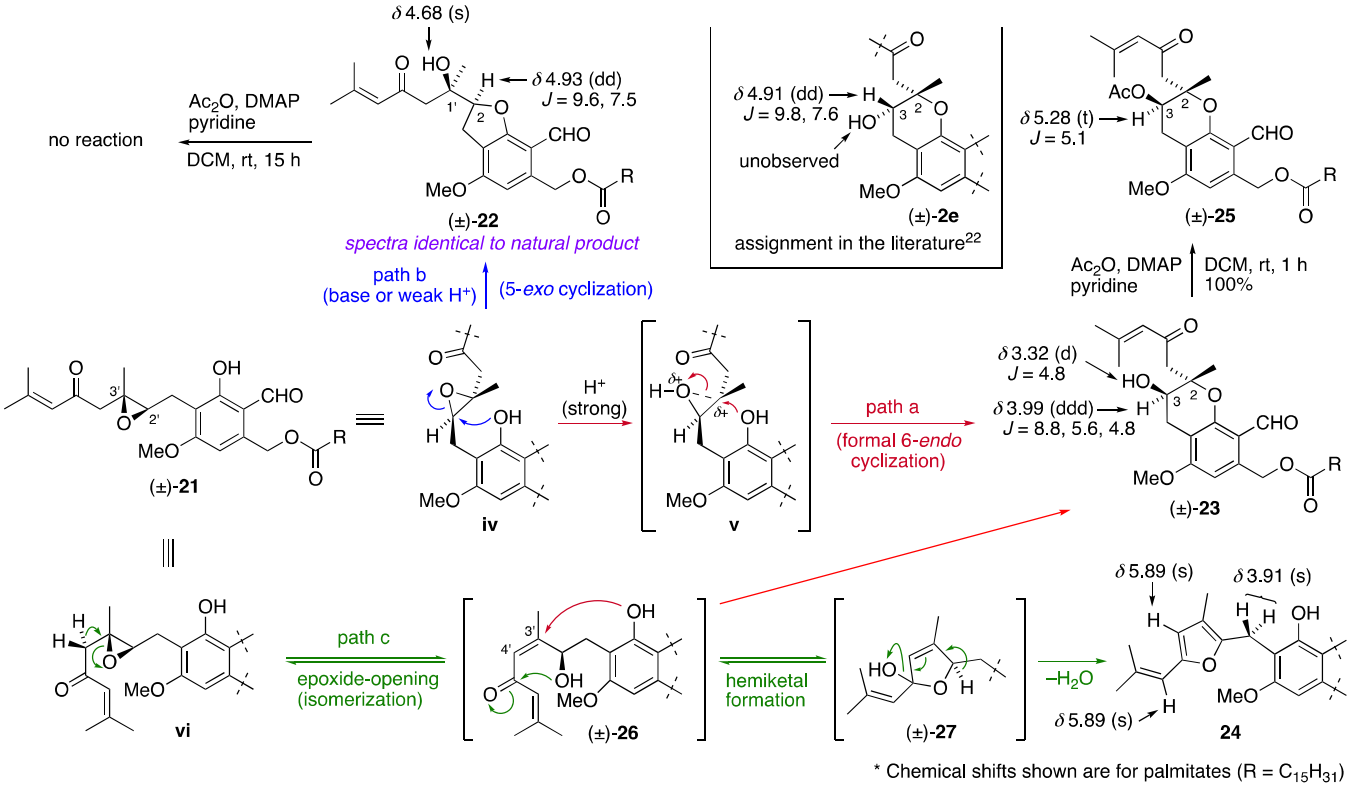
Scheme 6. Plausible Mechanisms to Produce (±)-22−24 and Spectroscopic Evidences to Determine Their Structures



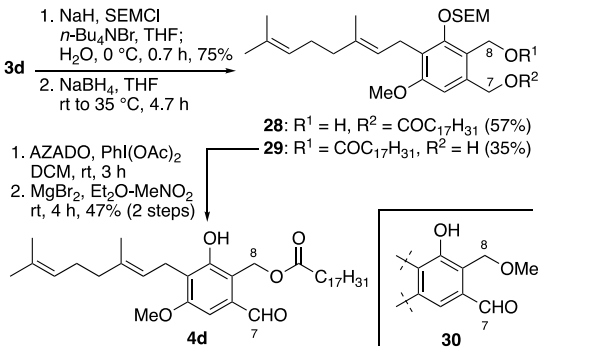
NMR spectrum,[54] and further derivatization to acetate (±)-**25**, in which the signal of C3−H was shifted to δ 5.28.[55] Along with a formal 6-*endo* cyclization, epoxide-opening (isomerization) to give the hydroxydienone (±)-**26** competed under strongly acidic conditions (path c). It is likely that resultant (±)-**26** was immediately converted to furan **24** via the hemiketal (±)-**27**. The epoxide-opening and dehydration processes seem fast under strong acidic conditions but relatively slow under weakly acidic or basic conditions. The labile intermediate detected by TLC in the case of benzoic acid or DBU is probably (±)-**26** [or its ketal form (±)-**27**]. It is also possible that the ring-closure product (±)-**23** is generated from enone (±)-**26** (or its trans stereoisomer at the C3′−C4′ olefin) via an intramolecular oxy Michael addition. However, the observation that (±)-**23** was obtained as a single diastereomer is thought to be the result reflecting a stereochemical requirement for epoxide-opening in conformation **v**.

In contrast to ring-closure processes under strong acidic conditions, 5-*exo* cyclization occurred smoothly under basic conditions with DBU (path b). The major product (±)-**22** was found to possess a tertiary alcohol whose signal appeared at δ 4.68 as a singlet in $CDCl_3$. Importantly, the appearance of the peak was highly dependent on the purity and concentration of the sample, and in many cases taking an NMR spectrum of (±)-**22**, the hydroxyl signal was not observed. We speculate that the signal of OH in the structural assignment of the natural product was not observed previously.[22] Additional evidence justifying the structure of (±)-**22** is the presence of a "dd" peak at δ 4.93 corresponding to C2−H. It is reasonable to consider that the chemical shift of C2−H in (±)-**22** is higher than that of C3−H in (±)-**2e** or (±)-**23** from textbook knowledge. Predictions of ^{13}C NMR chemical shifts by the ωB97X-D/6-31G* density functional model supported our proposed structure (Table S1 in the Supporting Information). In addition, acetylation of (±)-**22** did not progress at all, even after 15 h, probably due to the poor reactivity of the hindered tertiary alcohol.

Finally, under weakly acidic conditions using benzoic acid, it is proposed that 5-*exo* cyclization occurred faster than formal 6-*endo* cyclization due to the decreased cationic property at C3′. However, as compared to basic conditions, 5-*exo* cyclization with benzoic acid was much slower and the formation of the furan gradually competed over the reaction time.

After completing the total syntheses of natural products, we proceeded with the synthesis of a regioisomeric derivative of hericene D (**4d**), expecting that the neuroprotective activity would be enhanced by introducing a linoleate moiety at C8 (Scheme 7).[26] In our previous study, we showed that the phenolic silyl group readily migrated to the neighboring

Scheme 7. Synthesis of a Regioisomer of Hericene D (4d)



https://dx.doi.org/10.1021/acs.joc.0c02681
J. Org. Chem. 2021, 86, 2602−2620

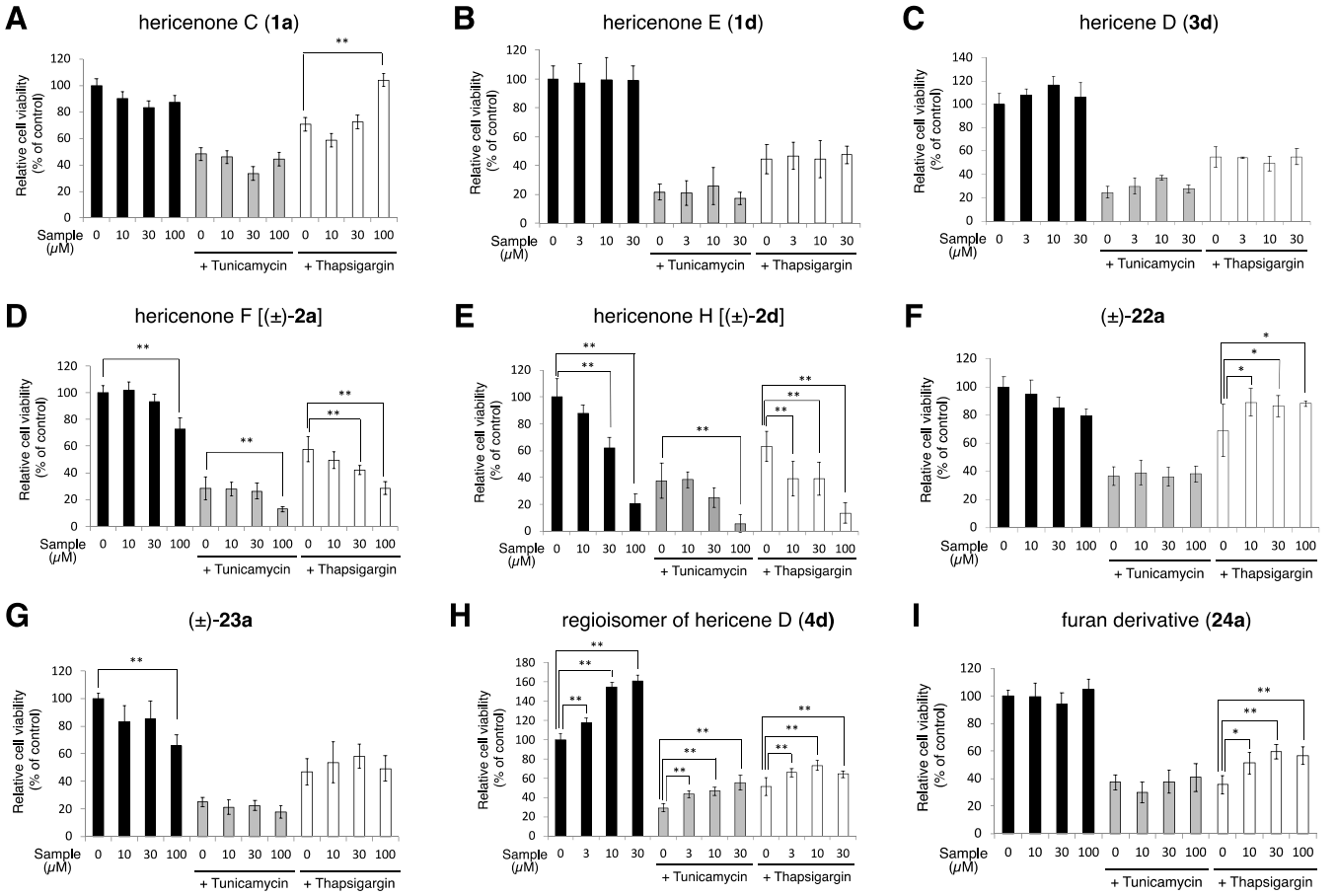


Figure 3. Neuroprotective effects of (A) hericenone C (**1a**), (B) hericenone E (**1d**), (C) hericene D (**3d**), (D) hericenone F [(±)-**2a**], (E) hericenone H [(±)-**2d**], (F) (±)-**22a**, (G) 3-hydroxyhericenone F [(±)-**23a**], (H) the regioisomer of hericene D (**4d**), and (I) the furan derivative **24a** against ER stress-dependent cell death. The cell viabilities were analyzed by MTT assays, and the values are represented as means ± SD of the relative percentage of surviving cells compared with the untreated cells (n = 6). We repeated each viability analysis at least three times and confirmed all of the data in the same condition showed similar trend and significance. * $p < 0.05$, ** $p < 0.01$, Tukey–Kramer multiple comparison tests.

primary hydroxyl group (C8–OH) under the hydride reduction conditions.[26] Therefore, we anticipated that the C7-ester group would migrate to the C8–OH in a similar way. After a short screening of conditions using **8a** and its derivatives as model substrates, reduction with $NaBH_4$ in anhydrous THF was found effective for ester migration. Thus, the phenolic hydroxyl group of hericene D (**3d**)[14] was protected as its 2-(trimethylsilyl)ethoxy methyl (SEM) ether[56] and subjected to reduction with $NaBH_4$ in anhydrous THF (Scheme 7). The ester-migration occurred gradually at 35 °C and gave the products **28** and **29** in 57 and 35% yields, respectively. It should be noted that the migration did not go to completion and a partial cleavage of ester to give a diol competed over the reaction time. After separation of regioisomers by flash chromatography, the C7-alcohol **29** was oxidized with AZADOL/PhI(OAc)$_2$,[57] to afford the corresponding C7-aldehyde. The last step, the removal of the SEM group, turned out to be more problematic than expected. The conventional method using n-Bu_4NF in the presence of N,N'-dimethylpropyleneurea[58] resulted in mostly decomposition. When methanolic HCl or its equivalent was used, benzyl methyl ether **30** was obtained as the major product. It is likely that the compound **30** was generated by methanol addition to the o-quinone methide intermediate produced by elimination of the ester moisty.[25] In the event, the desired product **4d** was obtained by applying the Hoffmann's conditions using excess $MgBr_2$ in Et_2O-$MeNO_2$[59] in 47% yield for two steps.

With synthetic hericenones and their derivatives in hand, potential neuroprotective effects were assayed against murine neuroblastoma cell line (Neuro2a) as the neuronal model.[60] Cell protection against the ER stress-dependent cell death was evaluated by using tunicamycin (TM), an inhibitor of N-linked glycosylation in the ER, and thapsigargin (TG), an inhibitor of sarcoplasmic/endoplasmic reticulum Ca^{2+}-ATPase, as positive ER stressors.[61,62] In Figure 3, the effects of compounds (with various concentrations) on the cell viability in the presence or the absence of ER stressors are represented.

Moderate neuroprotection against TG-induced ER stress was observed for hericenone C (**1a**) possessing a palmitoyl ester (chart **A**), while the stearoyl and oleoyl derivatives (**1b** and **1c**) did not show significant neuroprotection.[63] The neuroprotective activity of **1a** was comparable to that of hericenes A–C (**3a**–**3c**),[26] suggesting that the C5′-ketone was not the major determinant in neuroprotection. Hericenone E (**1d**) and hericene D (**3d**) with a linoleoyl group tended to slightly improve cell viability at low concentrations for TM-induced ER stress, but no significant differences were observed (charts **B** and **C**). Hericenones F [(±)-**2a**] and H [(±)-**2d**], the six-membered cyclized derivatives of **1a** and **1d**, showed toxicity rather than protective activity (charts **D** and **E**). It is

https://dx.doi.org/10.1021/acs.joc.0c02681
J. Org. Chem. 2021, 86, 2602–2620

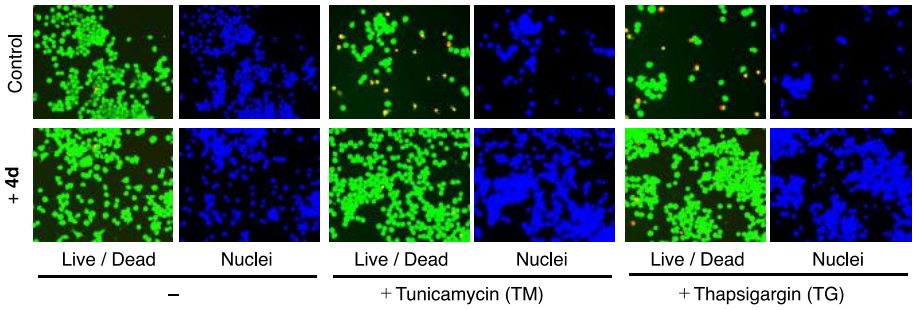


Figure 4. Cell staining after treatment with **4d** (30 μM) or without **4d** under the stress-free or the ER stress conditions. Living cells (green), dead cells (red), and nuclei (blue) were stained by Calcein-AM, propidium iodide, and Hoechst 33342, respectively. Stained cells were observed under a fluorescent microscope.

likely that the linoleoyl ester has more impact in cell viability than the palmitoyl ester. It is noteworthy that the actual 3-hydroxyhericenone F [(±)-**23**] was almost inactive for neuroprotection (chart **G**), while the toxicity observed for hericenone F was weakened by the presence of a hydroxyl group at C3 (chart **G** vs **D**). Importantly, the 5-*exo* cyclization product [(±)-**22a**], i.e., the putative 3-hydroxyhericenone F, showed significant protective effect against TG-induced neuronal cell death at more than 10 μM (chart **F**). This observation is consistent with a previous report,[22] although there was no significant neuroprotection against TM-induced cell death in the present study. Of interest is that more effective neuroprotection was observed for the regioisomer of hericene D (**4d**) possessing a linoleoyl ester at C8 (chart **H**). The protection of **4d** was significant even at 3 μM against both stressors. To demonstrate if **4d** actually protected cells from stress-induced cell death, we performed live/dead cell staining with Calcein-AM (live) and propidium iodide (dead). As shown in Figure 4, TM and TG reduced the total cell number and increased the ratio of red dead cells. The addition of **4d** clearly increased the ratio of green live cells, which indicates that **4d** actually protected the cells from ER stress. Unexpectedly, **4d** increased the cell number even in the absence of stressors (chart **H** in Figure 3). This finding suggests that **4d** possesses a cell growth-inducing function distinct from other hericenones.[64] The furan derivative **24a** was also assayed and found to be effective for neuroprotection against TG-induced ER stress at more than 10 μM (chart **I**).

■ CONCLUSIONS

In summary, the first total syntheses of hericenones C−H and "putative 3-hydroxyhericenone F", all of which possess a ketone and fatty ester functional groups, were achieved. Highlights of the synthesis include the straightforward construction of the resorcinol core and geranyl side chain, assembly of the natural product skeleton by sequential *O*-geranylation and a clay/zeolite-mediated 1,3-rearrangement reaction, and a biomimetic cyclization to form a variety of bicyclic natural hericenones and their congeners. The structure of "putative 3-hydroxyhericenone F (**2e**)" was revised as the 5-*exo* cyclization product of epoxyhericenone C, i.e., (±)-**22**, through in-depth analyses of cyclization reactions in addition to NMR spectroscopic studies. Here, we propose to name the compound (±)-**22** as "hericenone Z".[65] Although the natural product has been reported as a racemate on the basis of the CD spectrum data,[22] reassessment is desirable because generally enzymatic epoxidation takes place in an asymmetric manner. To obtain information on the biological functions of geranyl-resorcinols derived from *H. erinaceus*, the neuroprotective effects against ER stress-dependent cell death were systematically evaluated and primary structure−activity relationships were clarified. Among the compounds tested, the linoleate-containing derivative **4d** displayed the most potent neuroprotection, which indicates that the optimal positioning of the linoleate group is important for activity. More detailed mechanistic studies are ongoing in our group to understand the unprecedented ability of **4d** to enhance cell viability in unstressed cells.

■ EXPERIMENTAL SECTION

General Techniques. All reactions utilizing air- or moisture-sensitive reagents were performed under an atmosphere of argon. Commercially available dry solvents were used for THF and DCM. MeCN was dried over molecular sieves 4 Å before use. Pyridine and Et_3N were distilled from CaH_2 and dried over molecular sieves 4 Å before use. Commercially available NaH was dried under reduced pressure after the removal of mineral oil by washing with anhydrous *n*-hexane. Commercially available *m*-CPBA (purity: >69%) was dissolved in DCM to a concentration of 0.1 M and washed with 10% aqueous $NaHCO_3$ before epoxidation. Other chemicals were used as received. An oil bath was used for the heating reactions unless otherwise stated. Reactions were monitored by thin-layer chromatography (TLC) carried out on 0.25 mm silica gel plates (60-F254) that were analyzed by fluorescence upon 254 nm irradiation or by staining with *p*-anisaldeyde/AcOH/H_2SO_4/EtOH, $12MoO_3{\cdot}H_3PO_4$/EtOH, or $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O/H_2SO_4$ upon heating. The products were purified by either open chromatography on silica gel (spherical, neutral, 70−230 μm) or flash chromatography on silica gel (spherical, neutral, 40−50 μm) and, if necessary, HPLC equipped with a prepacked column with an eluent of *n*-hexane/EtOAc. NMR spectra were recorded with a 300 MHz (^{1}H: 300 MHz, ^{13}C: 75 MHz) or a 400 MHz (^{1}H: 400 MHz, ^{13}C: 101 MHz) spectrometer and referenced to the solvent peak at 7.26 ppm (^{1}H) and 77.16 ppm (^{13}C) for $CDCl_3$, or 7.16 ppm (^{1}H) and 128.06 ppm (^{13}C) for C_6D_6. Splitting patterns are indicated as follows: br, broad; s, singlet; d, doublet; t, triplet; q, quartet; quint, quintet; sept, septet; m, multiplet. Assignments of ^{1}H NMR chemical shifts for several compounds are based on ^{1}H−^{1}H COSY spectra. Infrared spectra were recorded with a FT-IR spectrometer and reported as wavenumber (cm^{-1}). High-resolution atmospheric pressure chemical ionization (APCI) and electron spray ionization (ESI) mass spectra were recorded with an Orbitrap analyzer in positive or negative ion mode. The carbon numbering of all synthetic compounds corresponds to that of natural products.

3,5-Dimethoxybenzyl Palmitate (6a).[39] 3,5-Dimethoxybenzyl alcohol (**5**) (4.65 g, 27.7 mmol) and DMAP (169 mg, 1.38 mmol) were placed in a well-dried 300 mL two-necked round-bottom flask, which was dissolved in 4-MeTHP (70 mL). Pyridine (6.7 mL, 82.8 mmol) and palmitoyl chloride (9.2 mL, 30 mmol) were added, and the mixture was stirred at room temperature for 30 min. The reaction

https://dx.doi.org/10.1021/acs.joc.0c02681
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mixture was quenched by the addition of saturated aqueous $NaHCO_3$ and extracted with EtOAc (2 × 25 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residual solid was recrystallized from *n*-hexane to give ester **6a** (10.8 g, 26.6 mmol, 96%) as a colorless solid. The mother liquid was further purified by flash chromatography (*n*-hexane → *n*-hexane/EtOAc, 30/1 v/v) to give **6a** (447 mg, 1.10 mmol, 4%). **6a**: colorless solid; mp 80−83 °C (*n*-hexane); ^{1}H NMR (400 MHz, $CDCl_3$) δ 6.49 (2H, d, *J* = 2.4 Hz), 6.41 (1H, t, *J* = 2.4 Hz), 5.05 (2H, s), 3.79 (6H, s), 2.36 (2H, t, *J* = 7.6 Hz), 1.68−1.60 (2H, m), 1.33−1.21 (24H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 173.8, 161.0, 138.5, 106.0, 100.2, 66.1, 55.5, 34.5, 32.1, 29.84, 29.82, 29.80, 29.75, 29.61, 29.51, 29.42, 29.31, 25.1, 22.8, 14.3; FT-IR (film on ZnSe) 3000, 2925, 2853, 1739, 1610, 1599, 1466, 1430, 1350, 1322, 1298, 1206, 1157, 1071 cm^{-1}; HRMS (ESI+): *m/z* calcd for $C_{25}H_{43}O_4$ [M + H]$^+$ 407.3156, found 407.3141.

2-Formyl-3,5-dimethoxybenzyl Palmitate (7a).[39] $POCl_3$ (4.2 mL, 45 mmol) was added to DMF (7.0 mL, 91 mmol) at 0 °C. The mixture was warmed to room temperature and stirred for 40 min. Ester **6a** (9.12 g, 22.4 mmol) was added, and the resulting mixture was stirred for 4 h at 80 °C. The reaction mixture was diluted with DCM and washed with saturated aqueous $NaHCO_3$ (120 mL). The aqueous layer was extracted with DCM (3 × 50 mL), and the combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated to give a substantially pure aldehyde **7a** (9.61 g, 22.1 mmol, 98%), which was used to the next reaction without further purification. **7a**: light brownish solid; mp 83−85 °C (*n*-hexane); ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.44 (1H, d, *J* = 0.6 Hz), 6.60 (1H, dd, *J* = 2.4, 0.6 Hz), 6.40 (1H, d, *J* = 2.4 Hz), 5.51 (2H, s), 3.89 (3H, s), 3.87 (3H, s), 2.41 (2H, t, *J* = 7.5 Hz), 1.68 (2H, quint, *J* = 7.5 Hz), 1.31−1.18 (24H, m), 0.87 (3H, t, *J* = 6.7 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 189.9, 173.3, 165.4, 165.3, 142.7, 115.9, 104.5, 96.6, 64.5, 56.0, 55.6, 34.5, 32.0, 29.82, 29.80, 29.78, 29.73, 29.60, 29.49, 29.43, 29.34, 25.2, 22.8, 14.2; FT-IR (film on ZeSe) 2999, 2924, 2853, 1738, 1599, 1464, 1429, 1350, 1321, 1298 cm^{-1}; HRMS (ESI+): *m/z* calcd for $C_{26}H_{43}O_5$ [M + H]$^+$ 435.3105, found 435.3089.

2-Formyl-3-hydroxy-5-methoxybenzyl Palmitate (8a).[39] BBr_3 (1.0 M solution in DCM, 5.9 mL, 5.9 mmol) was added to a solution of aldehyde **7a** (2.32 g, 5.34 mmol) in MeCN (140 mL) at 0 °C. The mixture was stirred for 1.5 h at room temperature and quenched by the addition of saturated aqueous $NaHCO_3$ (15 mL) and water (30 mL). The resulting mixture was extracted with EtOAc (1 × 150 mL, 1 × 20 mL), and the combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residual solid was recrystallized from *n*-hexane to give phenol **8a** (2.23 g, 5.29 mmol, 99%). **8a**: colorless solid; mp 81−83 °C (*n*-hexane/EtOAc); ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.46 (1H, s, OH), 10.06 (1H, s), 6.51 (1H, d, *J* = 2.4 Hz), 6.40 (1H, d, *J* = 2.4 Hz), 5.30 (2H, s), 3.85 (3H, s), 2.34 (2H, t, *J* = 7.4 Hz), 1.62 (2H, m), 1.31−1.21 (24H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 192.7, 173.3, 166.9, 166.6, 140.6, 112.4, 110.9, 100.8, 62.7, 55.9, 34.3, 32.1, 29.84, 29.81, 29.80, 29.78, 29.72, 29.57, 29.50, 29.35, 29.24, 25.0, 22.8, 14.3; FT-IR (film on ZeSe) 2918, 2851, 1732, 1622, 1327, 1281, 1206, 1155 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{25}H_{41}O_5$ [M + H]$^+$ 421.2949; found: 421.2943.

3,5-Dimethoxybenzyl Stearate (6b). Pyridine (1.1 mL, 14 mmol) and stearoyl chloride (1.69 mL, 5.00 mmol) were added to a solution of alcohol **5** (758 mg, 4.51 mmol) and DMAP (27.6 mg, 0.226 mmol) in 4-MeTHP (12 mL). After 30 min at room temperature, the reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ and extracted with EtOAc (2 × 20 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residual solid was recrystallized from *n*-hexane to give ester **6b** (1.51 g, 3.49 mmol, 77%). The mother liquid was further purified by flash chromatography (*n*-hexane/EtOAc, 20/1 v/v) to give ester **6b** (441 mg, 1.02 mmol, 23%). **6b**: colorless solid; mp 66−69 °C (*n*-hexane); ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.49 (2H, d, *J* = 2.4 Hz), 6.41 (1H, t, *J* = 2.4 Hz), 5.05 (2H, s), 3.79 (6H, s), 2.36 (2H, t, *J* = 7.5 Hz), 1.71−1.57 (2H, m), 1.32−1.19 (28H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 173.7, 161.0, 138.5, 106.0, 100.2, 66.1, 55.48, 55.46, 34.5, 32.1, 29.85, 29.82, 29.81, 29.75, 29.62, 29.51, 29.42, 29.31, 25.1, 22.8, 14.3; FT-IR (film on ZnSe) 2955, 2916, 2847, 1732, 1605, 1470, 1420, 1362, 1296, 1211, 1192, 1173, 1061 cm^{-1}; HRMS (ESI+): *m/z* calcd for $C_{27}H_{47}O_4$ [M + H]$^+$ 435.3469, found 435.3466.

2-Formyl-3,5-dimethoxybenzyl Stearate (7b). $POCl_3$ (830 μL, 9.02 mmol) was added to DMF (1.4 mL, 18 mmol) at 0 °C. The mixture was warmed to room temperature and stirred for 30 min. The solution was added to ester **6b** (1.96 g, 4.51 mmol), and the resulting mixture was stirred for 5 h at 80 °C. After the reaction mixture was cooled to 0 °C, it was diluted with *n*-hexane/EtOAc (5/1 v/v, 50 mL) and quenched by the addition of water (3 mL). The precipitate was collected by filtration, and the filtrate was concentrated. The collected solid and residue of the filtrate were combined, which was again extracted with DCM (3 × 50 mL) after the addition of water. The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residual solid was repeatedly recrystallized from *n*-hexane to give aldehyde **7b** (1.87 g, 4.04 mmol, 90%). **7b**: light brownish solid; mp 78−80 °C (*n*-hexane); ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.45 (1H, d, *J* = 0.6 Hz), 6.60 (1H, dd, *J* = 2.4, 0.6 Hz), 6.40 (1H, d, *J* = 2.4 Hz), 5.52 (2H, s), 3.90 (3H, s), 3.88 (3H, s), 2.42 (2H, t, *J* = 7.7 Hz), 1.68 (2H, quint, *J* = 7.4 Hz), 1.31−1.18 (28H, m), 0.87 (3H, t, *J* = 6.7 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 190.0, 173.3, 165.4, 165.3, 142.8, 115.9, 104.5, 96.7, 64.5, 56.1, 55.6, 34.5, 32.1, 29.85, 29.83, 29.81, 29.76, 29.63, 29.51, 29.45, 29.37, 25.2, 22.8, 14.3; FT-IR (film on ZeSe) 2916, 2851, 1728, 1667, 1597, 1470, 1458, 1431, 1300, 1215, 1192, 1153 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{28}H_{47}O_5$ [M + H]$^+$ 463.3418, found 463.3421.

2-Formyl-3-hydroxy-5-methoxybenzyl Stearate (8b). BBr_3 (1.0 M solution in DCM, 3.7 mL, 3.7 mmol) was added to a solution of aldehyde **7b** (1.54 g, 3.34 mmol) in MeCN (24 mL) at 0 °C. The mixture was warmed to room temperature and stirred for 1.5 h. The reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ and extracted with hot EtOAc (3 × 10 mL). The combined organic layer was washed with water, brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residual solid was recrystallized from *n*-hexane/EtOAc (5/1 v/v) to give phenol **8b** (1.49 g, 3.33 mmol, 100%). **8b**: ivory solid; mp 81−83 °C (*n*-hexane/EtOAc); ^{1}H NMR (300 MHz, $CDCl_3$) δ 12.46 (1H, s, OH), 10.05 (1H, s), 6.50 (1H, d, *J* = 2.7 Hz), 6.39 (1H, br d, *J* = 2.1 Hz), 5.29 (2H, s), 3.85 (3H, s), 2.34 (2H, t, *J* = 7.5 Hz), 1.61 (2H, m), 1.31−1.18 (28H, m), 0.87 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 192.7, 173.3, 166.8, 166.6, 140.6, 112.4, 110.9, 100.8, 62.7, 55.9, 34.3, 32.1, 29.84, 29.81, 29.78, 29.72, 29.58, 29.51, 29.36, 29.24, 25.0, 22.8, 14.3; FT-IR (film on ZeSe) 2920, 2851, 1732, 1620, 1582, 1470, 1423, 1389, 1327, 1281, 1231, 1204, 1157 cm^{-1}; HRMS (ESI+): *m/z* calcd for $C_{27}H_{45}O_5$ [M + H]$^+$ 449.3262, found 449.3262.

3,5-Dimethoxybenzyl Oleate (6c). Oleoyl chloride (purity >82%, 3.5 mL, >8.7 mmol) was added to a solution of alcohol **5** (1.34 g, 7.97 mmol), DMAP (49.1 mg, 0.401 mmol), and pyridine (1.9 mL, 24 mmol) in 4-MeTHP (20 mL). After 1 h, the reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ (10 mL) and extracted with *n*-hexane (2 × 20 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 100/1 → 30/1 v/v) to give ester **6c** (3.35 g, 7.75 mmol, 97%). **6c**: pale yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.49 (2H, d, *J* = 2.3 Hz), 6.41 (1H, t, *J* = 2.3 Hz), 5.36−5.32 (2H, m), 5.05 (2H, s), 3.79 (6H, s), 2.36 (2H, t, *J* = 7.5 Hz), 2.04−1.97 (4H, m), 1.70−1.60 (2H, m), 1.36−1.24 (20H, m), 0.88 (3H, t, *J* = 6.6 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 173.8, 161.1, 138.5, 130.2, 129.9, 106.0, 100.2, 66.1, 55.5, 34.5, 32.1, 29.92, 29.84, 29.68, 29.47, 29.32, 29.28, 29.26, 27.4, 27.3, 25.1, 22.8, 14.3; FT-IR (film on ZnSe) 3001, 2924, 2855, 1740, 1601, 1466, 1431, 1350, 1323, 1300, 1242, 1204, 1157, 1069 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{27}H_{45}O_4$ [M + H]$^+$ 433.3312, found 433.3312.

2-Formyl-3,5-dimethoxybenzyl Oleate (7c). $POCl_3$ (0.85 mL, 9.1 mmol) was added to DMF (1.5 mL, 19 mmol) at 0 °C. The

https://dx.doi.org/10.1021/acs.joc.0c02681
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solution was warmed to room temperature and stirred for 30 min. A solution of ester **6c** (2.00 g, 4.62 mmol) in DMF (4 mL) was added, and the resulting mixture was stirred for 6 h at 80 °C. The reaction mixture was diluted with *n*-hexane/EtOAc (5/1 v/v) (30 mL) and quenched by the addition of water (20 mL) at 0 °C. The resulting mixture was extracted with *n*-hexane/EtOAc (5/1 v/v) (4 × 30 mL), and the combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 10/1 v/v) to give aldehyde **7c** (1.76 g, 3.82 mmol, 83%). **7c**: pale yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.45 (1H, s), 6.60 (1H, br s), 6.40 (1H, d, *J* = 2.4 Hz), 5.52 (2H, s), 5.36−5.32 (2H, m), 3.90 (3H, s), 3.87 (3H, s), 2.42 (2H, t, *J* = 7.7 Hz), 2.04−1.96 (4H, m), 1.74−1.63 (2H, m), 1.39−1.24 (20H, m), 0.87 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 189.9, 173.3, 165.4, 165.3, 142.8, 130.2, 129.9, 115.9, 104.5, 96.6, 64.5, 56.1, 55.6, 34.5, 32.0, 29.91, 29.84, 29.66, 29.46, 29.33, 29.26, 27.4, 27.3, 25.2, 22.8, 14.2; FT-IR (film on ZeSe) 3005, 2928, 2855, 1740, 1674, 1605, 1578, 1458, 1435, 1331, 1304, 1285, 1238, 1204, 1153, 1088, 1053 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{28}H_{45}O_5$ [M + H]$^+$ 461.3262, found 461.3263.

2-Formyl-3-hydroxy-5-methoxybenzyl Oleate (8c). BBr_3 (1 M solution in DCM, 3.4 mL, 3.4 mmol) was added to a solution of aldehyde **7c** (1.38 g, 3.00 mmol) in MeCN (30 mL) at 0 °C. After 1 h at 0 °C, the mixture was warmed to room temperature and stirred for 1.5 h. The reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ (15 mL) and extracted with EtOAc (2 × 20 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 20/1 v/v) to give phenol **8c** (1.30 g, 2.92 mmol, 97%). **8c**: colorless solid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 12.46 (1H, s, OH), 10.05 (1H, s), 6.51 (1H, d, *J* = 2.4 Hz), 6.40 (1H, d, *J* = 2.4 Hz), 5.36−5.32 (2H, m), 5.30 (2H, s), 3.85 (3H, s), 2.34 (2H, t, *J* = 7.5 Hz), 2.04−1.97 (4H, m), 1.67−1.57 (2H, m), 1.36−1.24 (20H, m), 0.87 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$) δ 192.7, 173.3, 166.9, 166.6, 140.7, 130.2, 129.9, 112.4, 110.9, 100.9, 62.7, 55.9, 34.3, 32.1, 29.92, 29.81, 29.67, 29.47, 29.26, 29.22, 27.4, 27.3, 25.0, 22.8, 14.3; FT-IR (film on ZeSe) 3005, 2928, 2855, 1740, 1647, 1628, 1582, 1489, 1462, 1435, 1381, 1327, 1300, 1227, 1207, 1161 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{27}H_{43}O_5$ [M + H]$^+$ 447.3105, found 447.3094.

3,5-Dimethoxybenzyl Linoleate (6d). Linoleoyl chloride was prepared as follows. $(COCl)_2$ (1.85 mL, 21.6 mmol) and DMF (26 μL, 0.34 mmol) were added to a solution of linoleic acid (purity >88%, 4.91 g, >15.4 mmol) in DCM (30 mL). The mixture was stirred at room temperature for 3 h, and excess $(COCl)_2$ was removed by evaporation. Formation of acyl chloride was confirmed by the FT-IR spectrum, which showed the intense absorption at 1803 cm^{-1} (C═O stretching). The linoleoyl chloride, thus prepared, was dissolved in DCM (8 mL), which was added to a solution of alcohol **5** (2.86 g, 17.0 mmol), DMAP (94.3 mg, 0.772 mmol), and pyridine (3.8 mL, 47 mmol) in DCM (38 mL). After 1 h at room temperature, the reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ (25 mL) and extracted with *n*-hexane (2 × 40 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 30/1 v/v) to give ester **6d** (7.14 g, 16.6 mmol, 98%). **6d**: colorless oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 6.49 (2H, d, *J* = 2.4 Hz), 6.41 (1H, t, *J* = 2.4 Hz), 5.42−5.29 (4H, m), 5.05 (2H, s), 3.79 (6H, s), 2.77 (2H, br t, *J* = 6.6 Hz), 2.36 (2H, t, *J* = 7.4 Hz), 2.07−2.02 (4H, m), 1.68−1.61 (2H, m), 1.38−1.24 (14H, m), 0.89 (3H, t, *J* = 7.2 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 173.7, 161.0, 138.5, 130.4, 130.2, 128.2, 128.0, 106.0, 100.2, 66.1, 55.5, 34.4, 31.7, 29.72, 29.48, 29.30, 29.25, 29.24, 27.33, 27.32, 25.8, 25.1, 22.7, 14.2; FT-IR (film on ZnSe) 3006, 2922, 2856, 1736, 1599, 1464, 1431, 1350, 1298, 1244, 1205, 1159, 1070 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{27}H_{43}O_4$ [M + H]$^+$ 431.3156, found 431.3152.

2-Formyl-3,5-dimethoxybenzyl Linoleate (7d). $POCl_3$ (2.3 mL, 25 mmol) was added to DMF (4.2 mL, 48 mmol) at 0 °C. The mixture was warmed to room temperature and stirred for 30 min. A solution of ester **6d** (5.20 g, 12.1 mmol) in DMF (6 mL) was added, and the resulting mixture was stirred at 80 °C for 5 h. The reaction mixture was diluted with *n*-hexane/EtOAc (5/1 v/v) and quenched by the addition of water (30 mL) at 0 °C. The resulting mixture was extracted with *n*-hexane/EtOAc (5/1 v/v) (3 × 40 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 7/1 v/v) to give aldehyde **7d** (5.04 g, 11.0 mmol, 91%). **7d**: yellow oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.45 (1H, s), 6.60 (1H, br s), 6.41 (1H, br s), 5.52 (2H, s), 5.42−5.29 (4H, m), 3.91 (3H, s), 3.88 (3H, s), 2.77 (2H, t, *J* = 6.4 Hz), 2.42 (2H, t, *J* = 7.6 Hz), 2.07−2.02 (4H, m), 1.72−1.64 (2H, m), 1.39−1.25 (14H, m), 0.89 (3H, t, *J* = 6.6 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 189.9, 173.3, 165.4, 165.3, 142.8, 130.4, 130.2, 128.2, 128.0, 116.0, 104.6, 96.7, 64.5, 56.1, 55.6, 34.5, 31.7, 29.73, 29.48, 29.33, 29.32, 29.25, 27.3, 25.8, 25.2, 22.7, 14.2; FT-IR (film on ZeSe) 3009, 2928, 2855, 2785, 1744, 1670, 1605, 1578, 1458, 1435, 1331, 1304, 1285, 1238, 1204, 1153, 1088, 1053, 1011 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{28}H_{43}O_5$ [M + H]$^+$ 459.3105, found 459.3107.

2-Formyl-3-hydroxy-5-methoxybenzyl Linoleate (8d). BBr_3 (1 M solution in DCM, 3.4 mL, 3.4 mmol) was added to a solution of aldehyde **7d** (1.38 g, 3.01 mmol) in MeCN (30 mL) at 0 °C. After 1 h at 0 °C, the reaction mixture was warmed to room temperature and stirred for 1 h. The reaction mixture was quenched by the addition of saturated aqueous $NaHCO_3$ and extracted with EtOAc (2 × 20 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 20/1 v/v) to give phenol **8d** (1.27 g, 2.87 mmol, 95%). **8d**: pale yellow oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 12.45 (1H, s, OH), 10.06 (1H, s), 6.51 (1H, d, *J* = 2.4 Hz), 6.39 (1H, d, *J* = 2.4 Hz), 5.43−5.27 (4H, m), 5.30 (2H, s), 3.85 (3H, s), 2.76 (2H, t, *J* = 5.6 Hz), 2.34 (2H, t, *J* = 7.5 Hz), 2.08−2.00 (4H, m), 1.67−1.57 (2H, m), 1.38−1.24 (14H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 192.7, 173.2, 166.8, 166.6, 140.6, 130.3, 130.1, 128.2, 128.0, 112.4, 110.9, 100.8, 62.7, 55.9, 34.3, 31.6, 29.68, 29.47, 29.25, 29.20, 29.18, 27.32, 27.29, 25.7, 25.0, 22.7, 14.2; FT-IR (film on ZeSe) 3009, 2928, 2855, 1744, 1647, 1632, 1624, 1578, 1489, 1458, 1435, 1381, 1327, 1300, 1227, 1207, 1161, 1042 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{27}H_{41}O_5$ [M + H]$^+$ 445.2949, found 445.2940.

(*E*)-2-(1-Ethoxyethoxy)-6-hydroxy-4-methyl-2-(2-methylprop-1-en-1-yl)hex-4-enenitrile (11). K_2CO_3 (2.41 g, 17.4 mmol) was added to a solution of benzoate **10**[25] (5.87 g, 15.8 mmol) in MeOH (160 mL). After 14 h at room temperature, the reaction mixture was quenched by the addition of water (60 mL) and concentrated to remove most of MeOH. The residue was extracted with EtOAc (2 × 40 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by open chromatography (*n*-hexane/EtOAc, 10/1 → 2/1 v/v) to give alcohol **11** (4.08 g, 15.3 mmol, 97%) as an inseparable 4:1 diastereomeric mixture. **11**: pale yellow oil; ^{1}H NMR (400 MHz, C_6D_6) δ 5.64−5.57 (1H, m), 5.29 (0.2H, m), 5.05−4.96 (1.8H, m), 3.98−3.94 (2H, m), 3.62−3.52 (1.6H, m), 3.48−3.40 (0.2H, m), 3.30−3.22 (0.2H, m), 2.58 (0.8H, d, *J* = 14 Hz), 2.52 (0.2H, d, *J* = 14 Hz), 2.44 (0.8H, d, *J* = 14 Hz), 2.39 (0.2H, d, *J* = 14 Hz); 1.80 (0.6H, br s), 1.79 (2.4H, br s), 1.71 (2.4H, br s), 1.67 (0.6H, br s), 1.47 (0.6H, d, *J* = 1.2 Hz), 1.43 (2.4H, d, *J* = 1.2 Hz), 1.32 (0.6H, br d, *J* = 5.2 Hz), 1.23 (2.4H, br t, *J* = 7.0 Hz), 1.19 (2.4H, br d, *J* = 5.2 Hz), 1.06 (0.6H, br t, *J* = 7.0 Hz); ^{13}C{^{1}H} NMR (101 MHz, C_6D_6) δ 140.5 (major), 138.5 (minor), 132.0 (major), 131.9 (minor), 131.7 (major), 131.6 (minor), 124.2 (minor), 122.4 (major), 120.6 (major), 119.8 (minor), 98.7 (minor), 97.3 (major), 74.9 (minor), 73.8 (major), 61.3 (minor), 60.0 (major), 59.33 (major), 59.32 (minor), 51.2 (major), 50.4 (minor), 27.0 (major), 26.9 (minor), 21.2 (minor), 20.8 (major), 18.9 (major and minor), 18.1 (major), 18.0 (minor), 15.4 (minor), 15.2 (major); FT-IR (film on ZeSe) 3441, 2978, 2936, 2886, 1667, 1447, 1385, 1254, 1138, 1080, 1026, 1003 cm^{-1}; HRMS (APCI−): *m/z* calcd for $C_{15}H_{25}O_3NCl$ [M + Cl]$^-$ 302.1528, found 302.1542.

https://dx.doi.org/10.1021/acs.joc.0c02681
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(*E*)-6-Bromo-2-(1-ethoxyethoxy)-4-methyl-2-(2-methylprop-1-en-1-yl)hex-4-enenitrile (12). MsCl (1.3 mL, 17 mmol) was added to a solution of alcohol **11** (1.96 g, 7.33 mmol) and Et_3N (5.2 mL, 37 mmol) in THF (120 mL) at −78 °C. After 5 min, a solution of LiBr (3.30 g, 38.0 mmol) in THF (30 mL) was added over 10 min. The reaction mixture was warmed to 0 °C and stirred for 1 h. The mixture was quenched by the addition of water (50 mL) and extracted with *n*-hexane (2 × 30 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated to give bromide **12** (2.37 g, 7.17 mmol, 98%), which was too unstable in $CDCl_3$ to obtain pure spectra, and used to the next reaction without further purification.

General Procedure for *O*-Alkylation of 8a−8d with 12. NaH (435 mg, 18.1 mmol) was added to a solution of phenol **8a** (2.50 g, 5.97 mmol) and *n*-Bu_4NBr (2.30 g, 7.13 mmol) in toluene (90 mL) at 0 °C. A solution of crude **12** (2.37 g, 7.17 mmol) in toluene (30 mL) was then added, and the mixture was stirred at 50 °C for 1.5 h. Additional NaH (65.2 mg, 2.72 mmol) was added, and the stirring was continued for additional 0.5 h. The reaction mixture was quenched by the addition of water (40 mL) and extracted with *n*-hexane (4 × 40 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was quickly purified by flash chromatography (*n*-hexane/EtOAc, 20/1 → 10/1 v/v) to give the corresponding aryl ether (3.63 g, 5.42 mmol, 91%), which gradually decomposed in $CDCl_3$ and was immediately used to the next reaction. The aryl ether (3.00 g, 4.48 mmol) dissolved in MeOH (150 mL) was treated with *p*-TsOH·H_2O (85.2 mg, 0.448 mmol) at room temperature for 2 h. Et_3N (624 μL, 4.48 mmol) was added at 0 °C, and the mixture was stirred for 0.5 h at 0 °C. The reaction mixture was quenched by the addition of phosphate buffer (pH 7, 40 mL) and extracted with *n*-hexane/EtOAc (10/1 v/v) (4 × 30 mL). The combined organic layer was washed with 1 M aqueous HCl, brine, dried over anhydrous $MgSO_4$, filtered and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 10/1 → 5/1 v/v) to give ketone **9a** (2.18 g, 3.82 mmol, 85%).

(*E*)-3-((3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Palmitate (9a). Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.60 (1H, m), 6.41 (1H, d, *J* = 2.4 Hz), 6.10 (1H, sept, *J* = 1.2 Hz), 5.58 (1H, m), 5.52 (2H, s), 4.68 (2H, d, *J* = 6.2 Hz), 3.87 (3H, s), 3.16 (2H, s), 2.42 (2H, t, *J* = 7.5 Hz), 2.15 (3H, d, *J* = 1.2 Hz), 1.88 (3H, d, *J* = 1.2 Hz), 1.78 (3H, br s), 1.72−1.64 (2H, m), 1.36−1.23 (24H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 197.9, 190.1, 173.3, 165.1, 164.6, 157.1, 142.6, 136.2, 123.3, 122.8, 116.0, 104.6, 97.5, 65.6, 64.4, 55.6, 54.9, 34.5, 32.0, 29.77, 29.76, 29.73, 29.69, 29.56, 29.44, 29.38, 29.29, 27.8, 25.1, 22.8, 20.9, 17.3, 14.2; FT-IR (film on ZeSe) 2970, 2947, 2916, 2851, 1744, 1667, 1605, 1574, 1474, 1447, 1420, 1285, 1200, 1177, 1157 cm^{-1}; HRMS (ESI−): *m/z* calcd for $C_{35}H_{54}O_6Cl$ [M + Cl]$^-$ 605.3614, found 605.3624.

(*E*)-3-((3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Stearate (9b). This compound was obtained from **8b** and **12** in 69% yield for two steps. Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.60 (1H, m), 6.41 (1H, d, *J* = 2.4 Hz), 6.10 (1H, sept, *J* = 1.2 Hz), 5.58 (1H, m), 5.52 (2H, s), 4.68 (2H, d, *J* = 6.4 Hz), 3.87 (3H, s), 3.16 (2H, s), 2.42 (2H, t, *J* = 7.5 Hz), 2.15 (3H, d, *J* = 1.2 Hz), 1.88 (3H, d, *J* = 1.2 Hz), 1.78 (3H, br s), 1.72−1.65 (2H, m), 1.36−1.23 (28H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 198.0, 190.1, 173.4, 165.2, 164.6, 157.1, 142.7, 136.3, 123.4, 122.8, 116.1, 104.7, 97.6, 65.7, 64.5, 55.6, 54.9, 34.5, 32.1, 29.83, 29.81, 29.79, 29.74, 29.61, 29.50, 29.43, 29.35, 27.9, 25.2, 22.8, 21.0, 17.3, 14.3; FT-IR (film on ZeSe) 2974, 2947, 2916, 2851, 1744, 1667, 1605, 1574, 1474, 1447, 1285, 1200, 1177, 1157 cm^{-1}; HRMS (ESI−): *m/z* calcd for $C_{37}H_{58}O_6Cl$ [M + Cl]$^-$ 633.3927, found 633.3943.

(*E*)-3-((3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Oleate (9c). This compound was obtained from **8c** and **12** in 59% yield for two steps. Yellow oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.60 (1H, m), 6.41 (1H, d, *J* = 2.3 Hz), 6.10 (1H, sept, *J* = 1.3 Hz), 5.58 (1H, m), 5.52 (2H, s), 5.37−5.31 (2H, m), 4.68 (2H, d, *J* = 6.4 Hz), 3.87 (3H, s), 3.16 (2H, s), 2.42 (2H, t, *J* = 7.6 Hz), 2.15 (3H, *J* = 1.2 Hz), 2.04−1.98 (4H, m), 1.88 (3H, d, *J* = 1.6 Hz), 1.78 (3H, br s), 1.72−1.65 (2H, m), 1.37−1.24 (20H, m), 0.87 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 197.9, 190.1, 173.3, 165.1, 164.6, 157.1, 142.6, 136.3, 130.1, 129.8, 123.4, 122.8, 116.0, 104.7, 97.5, 65.7, 64.5, 55.6, 54.9, 34.5, 32.0, 29.86, 29.80, 29.62, 29.42, 29.31, 29.29, 29.21, 27.9, 27.31, 27.27, 25.1, 22.8, 20.9, 17.3, 14.2; FT-IR (film on ZeSe) 3005, 2928, 2855, 1740, 1670, 1601, 1574, 1443, 1381, 1300, 1281, 1234, 1200, 1157 cm^{-1}; HRMS (ESI−): *m/z* calcd for $C_{37}H_{56}O_6Cl$ [M + Cl]$^-$ 631.3771, found 631.3786.

(*E*)-3-((3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Linolate (9d). This compound was obtained from **8d** and **12** in 58% yield for two steps. Yellow oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.60 (1H, m), 6.41 (1H, d, *J* = 2.0 Hz), 6.10 (1H, sept, *J* = 1.2 Hz), 5.58 (1H, m), 5.52 (2H, s), 5.42−5.29 (4H, m), 4.68 (2H, d, *J* = 6.4 Hz), 3.87 (3H, s), 3.16 (2H, s), 2.77 (2H, br t, *J* = 6.4 Hz), 2.42 (2H, t, *J* = 7.5 Hz), 2.15 (3H, *J* = 1.2 Hz), 2.07−2.02 (4H, m), 1.88 (3H, d, *J* = 1.2 Hz), 1.78 (3H, br s), 1.72−1.65 (2H, m), 1.39−1.24 (14H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 198.0, 190.1, 173.3, 165.2, 164.6, 157.2, 142.7, 136.3, 130.3, 130.2, 128.2, 128.0, 123.4, 122.8, 116.0, 104.7, 97.5, 65.7, 64.5, 55.6, 54.9, 34.5, 31.6, 29.7, 29.47, 29.34, 29.31, 29.25, 27.9, 27.3, 25.7, 25.1, 22.7, 21.0, 17.3, 14.2; FT-IR (film on ZeSe) 3009, 2928, 2855, 1740, 1670, 1605, 1574, 1443, 1412, 1381, 1319, 1300, 1281, 1234, 1200, 1157 cm^{-1}; HRMS (ESI$^-$): *m/z* calcd for $C_{37}H_{54}O_6Cl$ [M + Cl]$^-$ 629.3614, found 629.3633.

General Procedure for O-Alkylation of 8b and 8d with Geranyl Bromide. NaH (225 mg, 9.37 mmol) was added to a solution of phenol **8d** (1.33 g, 2.99 mmol) and *n*-Bu_4NBr (289 mg, 0.897 mmol) in toluene (18 mL) at 0 °C. After 30 min at 0 °C, a solution of geranyl bromide (650 mg, 2.99 mmol) in toluene (2 mL) was added. The resulting mixture was stirred at room temperature for 30 min and warmed to 50 °C. After 30 min, the reaction mixture was quenched by the addition of saturated aqueous NH_4Cl at 0 °C and extracted with *n*-hexane (2 × 20 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated to give a substantially pure aryl ether **13d** (1.70 g, 2.92 mmol, 98%), which was used to the next reaction without further purification.

(*E*)-3-((3,7-Dimethylocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Stearate (13b). This compound was obtained from **8b** and geranyl bromide in 98% yield. The analytical sample was obtained by flash chromatography (*n*-hexane/EtOAc, 50/1 v/v). Colorless solid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.59 (1H, d, *J* = 2.4 Hz), 6.41 (1H, d, *J* = 2.4 Hz), 5.52 (2H, s), 5.47 (1H, br t, *J* = 6.8 Hz), 5.07 (1H, m), 4.62 (2H, d, *J* = 6.3 Hz), 3.86 (3H, s), 2.42 (2H, t, *J* = 7.6 Hz), 2.17−2.05 (4H, m), 1.74 (3H, br s), 1.81−1.54 (2H, m), 1.67 (3H, br s), 1.61 (3H, br s), 1.38−1.18 (28H, m), 0.88 (3H, t, *J* = 6.6 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 190.3, 173.4, 165.1, 164.8, 142.6, 142.2, 132.1, 123.7, 118.7, 116.1, 104.5, 97.7, 65.9, 64.5, 55.6, 39.6, 34.5, 32.0, 29.82, 29.80, 29.78, 29.73, 29.60, 29.49, 29.43, 29.34, 26.3, 25.8, 25.2, 22.8, 17.8, 16.9, 14.2; FT-IR (film on ZeSe) 2916, 2851, 1744, 1670, 1605, 1574, 1470, 1443, 1385, 1327, 1281, 1234, 1200, 1173, 1157 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{37}H_{61}O_5$ [M + H]$^+$ 585.4514, found 585.4510.

3-(((*E*)-3,7-Dimethylocta-2,6-dien-1-yl)oxy)-2-formyl-5-methoxybenzyl Linoleate (13d). The analytical sample was obtained by flash chromatography (*n*-hexane/EtOAc, 50/1 v/v). Colorless oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.59 (1H, m), 6.40 (1H, d, *J* = 2.4 Hz), 5.51 (2H, s), 5.47 (1H, m), 5.41−5.28 (4H, m), 5.07 (1H, m), 4.62 (2H, d, *J* = 6.4 Hz), 3.86 (3H, s), 2.76 (2H, t, *J* = 6.6 Hz), 2.42 (2H, t, *J* = 7.6 Hz), 2.16−2.00 (8H, m), 1.74 (3H, d, *J* = 1.2 Hz), 1.68 (2H, m), 1.67 (3H, d, *J* = 0.8 Hz), 1.60 (3H, br s), 1.38−1.24 (14H, m), 0.88 (3H, t, *J* = 7.0 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 190.1, 173.2, 165.0, 164.7, 142.4, 142.1, 132.0, 130.2, 130.0, 127.9, 123.5, 118.5, 116.0, 104.3, 97.5, 65.8, 64.4, 55.4, 39.4, 34.3, 31.5, 29.57, 29.32, 29.18, 29.15, 29.09, 27.2, 26.2, 25.65, 25.58, 25.0, 22.6, 17.7, 16.7, 14.1; FT-IR (film on ZeSe) 3007, 2926, 2854, 1740, 1670, 1601, 1578, 1445, 1410, 1379, 1319, 1300, 1281, 1236,

https://dx.doi.org/10.1021/acs.joc.0c02681
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1198, 1155 cm^{-1}; HRMS (ESI+): *m/z* calcd for $C_{37}H_{57}O_5$ $[M + H]^+$ 581.4201, found 581.4206.

General Procedure for O → C Rearrangement Promoted by Molecular Sieves 13X (Table 1, Entry 9). Well-ground molecular sieves 13X (5.09 g) were placed in a two-necked round-bottom flask, which was dried by a heat gun under reduced pressure. A solution of aryl ether **9d** (1.70 g, 2.92 mmol) in toluene (30 mL) was added at room temperature, and the mixture was stirred for 5 h at 100 °C. The reaction mixture was filtered through a short Celite column with EtOAc as an eluent. The filtrate was concentrated and purified by flash chromatography (*n*-hexane → *n*-hexane/EtOAc, 100/1 → 80/1 → 60/1 v/v) to give a 4.1:1 mixture of hericene D (**3d**) and phenol **8d** (545 mg) and a 1:1.5 mixture of **3d** and **8d** (118 mg). The calculated yield of **3d** was 34% (591 mg, 1.02 mmol) in total. The analytical sample was obtained by HPLC equipped with a prepacked column [Mightysil, Si 60,250−20 (5 μm)] using 12% EtOAc in *n*-hexane as an eluent.

4-((*E*)-3,7-Dimethylocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Stearate (3b, Hericene C).[26] Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.37 (1H, s), 10.10 (1H, s), 6.52 (1H, s), 5.32 (2H, s), 5.16 (1H, m), 5.05 (1H, m), 3.91 (3H, s), 3.34 (2H, d, *J* = 7.2 Hz), 2.33 (2H, t, *J* = 7.6 Hz), 2.08−2.01 (2H, m), 1.98−1.92 (2H, m), 1.76 (3H, d, *J* = 0.8 Hz), 1.63 (3H, d, *J* = 1.2 Hz), 1.62 (2H, m), 1.56 (3H, br s), 1.31−1.22 (28H, m), 0.88, (3H, t, *J* = 7.0 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 193.3, 173.4, 163.6, 163.0, 138.5, 135.9, 131.4, 124.5, 121.3, 118.2, 113.0, 105.7, 63.1, 56.0, 39.9, 34.4, 32.1, 29.85, 29.82, 29.81, 29.79, 29.73, 29.59, 29.52, 29.37, 29.25, 26.8, 25.8, 25.0, 22.8, 21.5, 17.8, 16.2, 14.3.

4-((*E*)-3,7-Dimethylocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Linoleate (3d, Hericene D). Colorless oil; ^{1}H NMR (300 MHz, $CDCl_3$) δ 12.37 (1H, s, OH), 10.10 (1H, s, CHO), 6.52 (1H, s, H6), 5.42−5.26 (4H, m, H9″, H10″, H12″, H13″), 5.32 (2H, s, H7), 5.16 (1H, br t, *J* = 7.2 Hz, H2′), 5.05 (1H, m, H6′), 3.91 (3H, s, OMe), 3.34 (2H, d, *J* = 7.2 Hz, H1′), 2.76 (2H, br t, *J* = 6.0 Hz, H11″), 2.33 (2H, t, *J* = 7.5 Hz, H2″), 2.08−1.92 (8H, m, H4′, H5′, H8″, H14″), 1.76 (3H, d, *J* = 1.2 Hz, H9′), 1.70−1.55 (2H, m, H3″), 1.63 (3H, d, *J* = 1.2 Hz, H8′), 1.56 (3H, br s, H10′), 1.37−1.23 (14H, m, H4″−H7″, H15″−H17″), 0.88 (3H, t, *J* = 6.9 Hz, H18″); $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$) δ 193.2, 173.3, 163.6, 163.0, 138.5, 135.9, 131.4, 130.4, 130.1, 128.2, 128.0, 124.5, 121.3, 118.2, 113.0, 105.7, 63.1, 56.0, 39.9, 34.3, 31.7, 29.70, 29.48, 29.26, 29.22, 27.34, 27.31, 26.82, 25.81, 25.76, 25.0, 22.7, 21.5, 17.8, 16.2, 14.2; HRMS (APCI+): *m/z* calcd for $C_{37}H_{57}O_5$ $[M + H]^+$ 581.4201, found 581.4203.

General Procedure for O → C Rearrangement Promoted by Montmorillonite K10 and Molecular Sieves 4 Å (Table 1, Entry 15). Well-ground molecular sieves 4 Å (10.0 g) were placed in a two-necked 500 mL round-bottom flask, which was dried by a heat gun under reduced pressure. Aryl ether **9a** (1.00 g, 1.76 mmol), DCE (100 mL), and montmorillonite K10 (5.00 g) were added at room temperature, and the mixture was stirred for 48 h at 50 °C. The reaction mixture was filtered through a short Celite column with EtOAc as an eluent. The filtrate was concentrated and purified by flash chromatography (*n*-hexane/EtOAc, 15/1 → 10/1 v/v) to give hericenone C (**1a**) (308 mg, 0.539 mmol, 33%).

(*E*)-4-(3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Palmitate (1a, Hericenone C). Pale yellow solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.38 (1H, s), 10.10 (1H, s), 6.53 (1H, s), 6.09 (1H, sept, *J* = 1.2 Hz), 5.32 (2H, s), 5.31 (1H, m), 3.91 (3H, s), 3.39 (2H, d, *J* = 7.2 Hz) 3.00 (2H, s), 2.33 (2H, t, *J* = 7.2 Hz), 2.12 (3H, d, *J* = 1.2 Hz), 1.84 (3H, d, *J* = 1.2 Hz), 1.77 (3H, br s), 1.61 (2H, qui, *J* = 7.2 Hz), 1.31−1.23 (24H, m), 0.87 (3H, t, *J* = 7.2 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 199.8, 193.3, 173.4, 163.6, 163.0, 155.7, 138.8, 130.5, 126.4, 122.9, 117.4, 113.0, 105.7, 63.1, 56.1, 55.7, 34.4, 32.1, 29.83, 29.81, 29.80, 29.78, 29.72, 29.58, 29.50, 29.36, 29.25, 27.8, 25.0, 22.8, 21.7, 20.8, 16.5, 14.3; FT-IR (film on ZeSe) 2924, 2855, 1740, 1686, 1624, 1578, 1501, 1466, 1412, 1381, 1312, 1285, 1254, 1223, 1165, 1123, 1018 cm^{-1}; HRMS (ESI-pos): *m/z* calcd for $C_{35}H_{55}O_6$ $[M + H]^+$ 571.3993, found 571.3983.

(*E*)-4-(3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Stearate (1b, Hericenone D). This compound was obtained in 31% yield (154 mg, 0.257 mmol) from **9b** (501 mg, 0.837 mmol) by treatment with montmorillonite K10 (2.50 g) and molecular sieves 4 Å (5.00 g) for 48 h at 50 °C. Pale yellow solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.38 (1H, s, OH), 10.10 (1H, s), 6.52 (1H, s), 6.09 (1H, sept, *J* = 1.2 Hz), 5.32 (2H, s), 5.31 (1H, m), 3.91 (3H, s), 3.39 (2H, d, *J* = 7.2 Hz) 3.00 (2H, s), 2.33 (2H, t, *J* = 7.2 Hz), 2.12 (3H, d, *J* = 1.2 Hz), 1.84 (3H, d, *J* = 1.2 Hz), 1.77 (3H, br s), 1.61 (2H, m), 1.31−1.23 (28H, m), 0.87 (3H, t, *J* = 7.2 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 199.7, 193.3, 173.4, 163.6, 163.0, 155.7, 138.8, 130.5, 126.4, 122.9, 117.4, 113.0, 105.7, 63.1, 56.1, 55.7, 34.4, 32.1, 29.84, 29.81, 29.80, 29.78, 29.73, 29.58, 29.50, 29.37, 29.25, 27.8, 25.0, 22.8, 21.7, 20.8, 16.5, 14.3; FT-IR (film on ZeSe) 2921, 2851, 1734, 1686, 1623, 1576, 1506, 1467, 1412, 1387, 1312, 1285, 1254, 1221, 1157, 1121 cm^{-1}; HRMS (ESI-pos): *m/z* calcd for $C_{37}H_{59}O_6$ $[M + H]^+$ 599.4306, found 599.4298.

(*E*)-4-(3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Oleate (1c, 5′-Oxohericene B). This compound was obtained in 30% yield (149 mg, 0.249 mmol) from **9c** (500 mg, 0.838 mmol) by treatment with montmorillonite K10 (2.50 g) and molecular sieves 4 Å (5.01 g) for 48 h at 50 °C. Pale yellow oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.38 (1H, s, OH), 10.11 (1H, s), 6.53 (1H, s), 6.09 (1H, sept, *J* = 1.2 Hz), 5.37−5.29 (3H, m), 5.32 (2H, s), 3.91 (3H, s), 3.40 (2H, d, *J* = 7.3 Hz) 3.00 (2H, s), 2.33 (2H, t, *J* = 7.5 Hz), 2.12 (3H, d, *J* = 1.2 Hz), 2.03−1.98 (4H, m), 1.84 (3H, d, *J* = 1.2 Hz), 1.78 (3H, br s), 1.65−1.58 (2H, m), 1.35−1.24 (20H, m), 0.88 (3H, t, *J* = 7.0 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 199.6, 193.2, 173.2, 163.6, 163.0, 155.5, 138.8, 130.4, 130.1, 129.8, 126.3, 122.9, 117.4, 113.0, 105.6, 63.0, 56.0, 55.6, 34.3, 32.0, 29.85, 29.75, 29.62, 29.41, 29.22, 29.17, 29.16, 27.8, 27.3, 27.2, 24.9, 22.8, 21.7, 20.7, 16.5, 14.2; FT-IR (film on ZeSe) 3003, 2926, 2855, 1741, 1682, 1634, 1622, 1576, 1504, 1464, 1416, 1384, 1312, 1285, 1254, 1223, 1167, 1123 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{37}H_{57}O_6$ $[M + H]^+$ 597.4150, found 597.4138.

(*E*)-4-(3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Linoleate (1d, Hericenone E). This compound was obtained in 28% yield (139 mg, 0.234 mmol) from **9d** (502 mg, 0.843 mmol) by treatment with montmorillonite K10 (2.50 g) and molecular sieves 4 Å (5.00 g) for 48 h at 50 °C. Pale yellow oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.38 (1H, s, OH), 10.10 (1H, s), 6.52 (1H, s), 6.09 (1H, sept, *J* = 1.3 Hz), 5.41−5.28 (5H, m), 5.32 (2H, s), 3.91 (3H, s), 3.39 (2H, d, *J* = 7.1 Hz), 3.00 (2H, s), 2.76 (2H, br t, *J* = 6.4 Hz), 2.33 (2H, t, *J* = 7.5 Hz), 2.12 (3H, *J* = 1.2 Hz), 2.07−1.99 (4H, m), 1.84 (3H, d, *J* = 1.2 Hz), 1.77 (3H, br s), 1.61 (2H, m), 1.37−1.24 (14H, m), 0.88 (3H, t, *J* = 7.0 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 199.7, 193.3, 173.3, 163.6, 163.0, 155.7, 138.8, 130.5, 130.4, 130.1, 128.2, 128.0, 126.4, 122.9, 117.4, 113.0, 105.7, 63.1, 56.1, 55.7, 34.3, 31.7, 29.70, 29.48, 29.27, 29.21, 27.8, 27.33, 27.30, 25.7, 25.0, 22.7, 21.7, 20.8, 16.5, 14.2; FT-IR (film on ZeSe) 3007, 2930, 2857, 1738, 1688, 1634, 1622, 1576, 1504, 1464, 1416, 1385, 1312, 1285, 1254, 1225, 1167, 1123 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{37}H_{55}O_6$ $[M + H]^+$ 595.3993, found 595.3977.

The O → C Rearrangement Promoted by Montmorillonite K10 and Molecular Sieves 5 Å. Well-ground molecular sieves 5 Å (6.26 g) were placed in a two-necked 200 mL round-bottom flask, which was dried by a heat gun under reduced pressure. Montmorillonite K10 (3.30 g) and a solution of aryl ether **9a** (655 mg, 1.15 mmol) in DCE (50 mL) were added at room temperature, and the mixture was stirred at 50 °C for 48 h. The reaction mixture was filtered through a short Celite column with EtOAc as an eluent. The filtrate was concentrated and purified by flash chromatography (*n*-hexane/EtOAc, 20/1 → 15/1 → 10/1 v/v) to give hericenone C (**1a**) (215 mg, 0.377 mmol, 33%), phenol **8a** (219 mg, 0.520 mmol, 45%), the *para*-rearrangement product **16a** (12.0 mg, 0.0210 mmol, 2%), trienone **17** (19.5 mg, 0.130 mmol, 11%), and an 1.3:1 mixture of trienones **17** and **18** (3.3 mg, calculated yield: **17**: 1%, **18**: 0.8%).

(*E*)-2-(3,7-Dimethyl-5-oxoocta-2,6-dien-1-yl)-6-formyl-5-hydroxy-3-methoxybenzyl Palmitate (16a). Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.63 (1H, s), 10.11 (1H, s), 6.43 (1H,

https://dx.doi.org/10.1021/acs.joc.0c02681
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s), 6.06 (1H, m), 5.35 (2H, s), 5.10 (1H, m), 3.86 (3H, s), 3.49 (2H, d, J = 6.9 Hz) 3.01 (2H, s), 2.31 (2H, t, J = 7.5 Hz), 2.13 (3H, d, J = 0.9 Hz), 1.86 (3H, d, J = 1.4 Hz), 1.78 (3H, br s), 1.59 (2H, m), 1.32−1.22 (24H, m), 0.88 (3H, t, J = 6.9 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 199.2, 193.9, 173.4, 165.2, 164.8, 156.1, 137.1, 130.3, 127.3, 123.3, 122.9, 112.7, 99.8, 58.2, 56.1, 55.4, 34.3, 32.1, 29.84, 29.80, 29.78, 29.72, 29.62, 29.58, 29.51, 29.37, 29.25, 27.9, 25.0, 24.4, 22.8, 20.9, 16.7, 14.3; FT-IR (film on ZeSe) 2924, 2855, 1740, 1686, 1632, 1466, 1443, 1377, 1323, 1285, 1219, 1161, 1134 cm^{-1}; HRMS (APCI+): m/z calcd for $C_{35}H_{55}O_6$ [M + H]$^+$ 571.3993, found 571.3986.

2-Methyl-6-methyleneocta-2,7-dien-4-one (17). Yellow oil; ^1H NMR (400 MHz, $CDCl_3$) δ 6.43 (1H, br dd, J = 17, 11 Hz), 6.16 (1H, sept, J = 1.2 Hz), 5.25 (1H, br s), 5.16 (1H, br d, J = 17 Hz), 5.11 (1H, br s), 5.10 (1H, br d, J = 11 Hz), 3.30 (2H, d, J = 0.8 Hz), 2.14 (3H, d, J = 1.2 Hz), 1.88 (3H, d, J = 1.2 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 198.2, 156.7, 140.7, 138.3, 122.5, 120.1, 115.1, 48.1, 27.9, 20.9; FT-IR (film on ZeSe) 3009, 2974, 2932, 2916, 1686, 1620, 1597, 1443, 1381, 1223, 1107, 1045 cm^{-1}; HRMS (APCI-pos): m/z calcd for $C_{10}H_{15}O$ [M + H]$^+$ 151.1117, found 151.1119.

(*E*)-2,6-Dimethylocta-2,5,7-trien-4-one (18).[66] The selected ^1H NMR signals obtained from the mixture of **17** and **18**: ^1H NMR (400 MHz, $CDCl_3$) δ 6.39 (1H, br dd, J = 16, 11 Hz), 6.13 (2H, br s), 5.63 (1H, d, J = 17 Hz), 5.40 (1H, d, J = 11 Hz), 2.26 (3H, d, J = 1.2 Hz), 2.19 (3H, d, J = 1.2 Hz), 1.90 (3H, d, J = 1.2 Hz).

General Procedure for the Acid-Induced Cyclization of 1a−1d. (+)-CSA (64.5 mg, 0.278 mmol) was added to a solution of **1a** (81.8 mg, 0.143 mmol) in DCE (7 mL). The mixture was stirred for 48 h at 70 °C. After concentration, the residue was purified by flash chromatography (n-hexane → n-hexane/EtOAc, 10/1 v/v) to give (±)-hericenone F [(±)-**2a**] (62.5 mg, 0.109 mmol, 76%).

Base-Induced Cyclization of 1b. DBU (5 μL, 0.033 mmol) was added to a solution of **1b** (10.0 mg, 0.0167 mmol) in DCE (2 mL). The mixture was stirred for 20 h at room temperature. The solution was directly subjected to flash chromatography (n-hexane → n-hexane/EtOAc, 10/1 v/v) to give (±)-hericenone G [(±)-**2b**] (10.0 mg, 0.0167 mmol, 100%).

(±)-((*R)-8-Formyl-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Palmitate [(±)-2a, Hericenone F].** Colorless solid; ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (1H, d, J = 0.8 Hz), 6.55 (1H, br s), 6.06 (1H, m), 5.51 (2H, br s), 3.88 (3H, s), 2.82 (1H, d, J = 14.0 Hz), 2.66 (1H, d, J = 14.0 Hz), 2.64 (2H, t, J = 6.8 Hz), 2.41 (2H, t, J = 7.4 Hz), 2.14 (3H, d, J = 1.2 Hz), 2.02 (1H, dt, J = 13.6, 6.8 Hz), 1.91 (1H, dt, J = 13.6, 6.8 Hz), 1.86 (3H, d, J = 1.2 Hz′), 1.69 (2H, quint, J = 7.4 Hz), 1.44 (3H, s), 1.36−1.21 (24H, m), 0.88 (3H, t, J = 7.2 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 198.1, 190.5, 173.4, 162.1, 158.4, 156.6, 139.8, 125.1, 115.9, 109.4, 101.0, 77.1, 64.7, 55.7, 52.6, 34.6, 32.1, 30.2, 29.84, 29.82, 29.80, 29.76, 29.62, 29.50, 29.47, 29.40, 27.9, 25.3, 24.7, 22.8, 21.0, 16.6, 14.3; FT-IR (film on ZeSe) 2924, 2855, 1740, 1670, 1605, 1574, 1462, 1400, 1377, 1300, 1215, 1157, 1115, 1042 cm^{-1}; HRMS (ESI+): m/z calcd for $C_{35}H_{55}O_6$ [M + H]$^+$ 571.3993, found 571.3989.

(±)-((*R)-8-Formyl-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Stearate [(±)-2b, Hericenone G].** In addition to the based-induced cyclization described above, this compound was obtained in 83% yield (18.0 mg, 0.0301 mmol) from **1b** (21.6 mg, 0.0361 mmol) by treatment with (±)-CSA (17.0 mg, 0.0732 mmol) for 48 h at 70 °C. Colorless solid; ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (1H, d, J = 0.4 Hz), 6.54 (1H, br s), 6.06 (1H, sept, J = 1.2 Hz), 5.50 (2H, br s), 3.88 (3H, s), 2.82 (1H, d, J = 14.0 Hz), 2.66 (1H, d, J = 14.0 Hz), 2.63 (2H, t, J = 6.8 Hz), 2.41 (2H, t, J = 7.6 Hz), 2.14 (3H, d, J = 1.2 Hz), 2.01 (1H, dt, J = 13.6, 6.8 Hz), 1.91 (1H, dt, J = 13.6, 6.8 Hz), 1.86 (3H, d, J = 1.2 Hz), 1.68 (2H, quint, J = 7.2 Hz), 1.44 (3H, s), 1.36−1.24 (28H, m), 0.88 (3H, t, J = 7.2 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 198.1, 190.5, 173.4, 162.1, 158.4, 156.6, 139.8, 125.1, 115.9, 109.3, 100.9, 77.1, 64.7, 55.7, 52.6, 34.6, 32.1, 30.2, 29.84, 29.82, 29.80, 29.76, 29.62, 29.51, 29.47, 29.39, 27.9, 25.3, 24.6, 22.8, 21.0, 16.6, 14.3; FT-IR (film on ZeSe) 2922, 2853, 1740, 1672, 1603, 1574, 1465, 1420, 1400, 1377, 1340, 1298, 1246, 1215, 1157, 1116, 1040 cm^{-1}; HRMS (ESI-pos): m/z calcd for $C_{37}H_{59}O_6$ [M + H]$^+$ 599.4306, found 599.4294.

(±)-((*R)-8-Formyl-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Oleate [(±)-2c].** This compound was obtained in 80% yield (25.7 mg, 0.0431 mmol) from **1c** (32.3 mg, 0.0541 mmol) by treatment with (±)-CSA (24.2 mg, 0.104 mmol) for 48 h at 70 °C. Pale yellow oil; ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (1H, br s), 6.54 (1H, br s), 6.06 (1H, sept, J = 1.2 Hz), 5.51 (2H, br s), 5.37−5.31 (2H, m), 3.88 (3H, s), 2.82 (1H, d, J = 14.0 Hz), 2.66 (1H, d, J = 14.0 Hz), 2.63 (2H, t, J = 6.8 Hz), 2.41 (2H, t, J = 7.5 Hz) 2.14 (3H, d, J = 1.2 Hz), 2.05−1.98 (5H, m), 1.91 (1H, dt, J = 13.6, 6.8 Hz), 1.86 (3H, d, J = 1.2 Hz), 1.73−1.64 (2H, m), 1.44 (3H, s), 1.37−1.24 (20H, m), 0.87 (3H, t, J = 6.8 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 198.1, 190.5, 173.4, 162.0, 158.4, 156.6, 139.8, 130.2, 129.8, 125.1, 115.8, 109.3, 100.9, 77.0, 64.7, 55.7, 52.6, 34.5, 32.0, 30.1, 29.89, 29.84, 29.66, 29.46, 29.36, 29.35, 29.25, 27.9, 27.34, 27.29, 25.2, 24.6, 22.8, 21.0, 16.6, 14.3; FT-IR (film on ZeSe) 3003, 2926, 2855, 1741, 1670, 1603, 1574, 1464, 1298, 1215, 1155, 1117 cm^{-1}; HRMS (APCI+): m/z calcd for $C_{37}H_{57}O_6$ [M + H]$^+$ 597.4150, found 597.4140.

(±)-((*R)-8-Formyl-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Linoleate [(±)-2d, Hericenone H].** This compound was obtained in 68% yield (20.8 mg, 0.0350 mmol) from **1d** (30.4 mg, 0.0511 mmol) by treatment with (±)-CSA (23.5 mg, 0.101 mmol) for 48 h at 70 °C. Colorless oil; ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (1H, br s), 6.54 (1H, br s), 6.06 (1H, sept, J = 1.2 Hz), 5.51 (2H, br s), 5.42−5.29 (4H, m), 3.88 (3H, s), 2.82 (1H, d, J = 14 Hz), 2.77 (2H, br t, J = 6.4 Hz), 2.66 (1H, d, J = 14 Hz), 2.63 (2H, t, J = 6.8 Hz), 2.41 (2H, t, J = 7.8 Hz) 2.14 (3H, d, J = 1.2 Hz), 2.07−2.01 (4H, m), 2.01 (1H, dt, J = 14, 7.2 Hz), 1.91 (1H, dt, J = 14, 7.0 Hz), 1.86 (3H, d, J = 1.2 Hz), 1.73−1.64 (2H, m), 1.44 (3H, s), 1.39−1.25 (14H, m), 0.88 (3H, t, J = 7.0 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 198.1, 190.5, 173.4, 162.0, 158.4, 156.7, 139.8, 130.4, 130.1, 128.2, 128.0, 125.1, 115.8, 109.3, 100.9, 77.0, 64.7, 55.7, 52.6, 34.5, 31.6, 30.1, 29.73, 29.47, 29.36, 29.34, 29.25, 27.9, 27.324, 27.315, 25.7, 25.2, 24.6, 22.7, 21.0, 16.6, 14.2; FT-IR (film on ZeSe) 3007, 2930, 2857, 1738, 1674, 1603, 1574, 1464, 1377, 1298, 1215, 1157, 1117 cm^{-1}; HRMS (APCI+): m/z calcd for $C_{37}H_{55}O_6$ [M + H]$^+$ 595.3993, found 595.3981.

Tracking of the Reaction Progress for Cyclization of Hericenone D (1b) with (+)-CSA (Table 2, Entry 1). (+)-CSA (7.8 mg, 0.034 mmol) was added to a solution of **1b** (10.3 mg, 0.0172 mmol) in DCE (2 mL). After 20 h at room temperature, the solution was directly subjected to flash chromatography (n-hexane/EtOAc, 10/1 v/v) to give an 89:9:2 mixture of **1b**, **20b**, and **2b** (10.3 mg). The product was dissolved in DCE (1 mL), which was treated with (+)-CSA (7.9 mg, 0.034 mmol) for 3 h at 50 °C. The solution was directly subjected to flash chromatography (n-hexane/EtOAc, 10/1 v/v) to give a 75:21:4 mixture of **1b**, **20b**, and **2b** (9.1 mg). The product was again dissolved in DCE (1 mL), which was treated with (+)-CSA (7.9 mg, 0.034 mmol) for 16 h at 50 °C. The solution was directly subjected to flash chromatography (n-hexane/EtOAc, 10/1 v/v) to give a 32:27:41 mixture of **1b**, **20b**, and **2b** (8.2 mg).

Tracking of the Reaction Progress for Cyclization of Hericenone D (1b) with DBU (Table 2, Entry 2). DBU (5 μL, 0.033 mmol) was added to a solution of **1b** (10.3 mg, 0.0172 mmol) in DCE (1 mL) at 0 °C. The mixture was stirred for 1.5 h at 0 °C and directly subjected to flash chromatography (n-hexane/EtOAc, 10/1 v/v) to give a 60:9:31 mixture of **1b**, **20b**, and **2b** (8.6 mg).

(*E*)-4-(3,7-Dimethyl-5-oxoocta-3,6-dien-1-yl)-2-formyl-3-hydroxy-5-methoxybenzyl Stearate (20b). The selected ^1H NMR signals of **20b** obtained from the mixture of **1b**, **20b**, and **2b**: ^1H NMR (400 MHz, $CDCl_3$) δ 12.39 (1H, s), 10.11 (1H, s), 6.52 (1H, s), 6.03 (1H, m), 6.01 (1H, m), 5.32 (2H, s), 3.90 (3H, s), 2.81 (2H, m), 2.33 (2H, t, J = 7.6 Hz), 2.28 (2H, m), 2.23 (3H, d, J = 1.0 Hz), 2.15 (3H, d, J = 1.0 Hz, H9′), 1.88 (3H, d, J = 1.0 Hz), 1.61 (2H, m), 1.35−1.21 (28H, m), 0.87 (3H, t, J = 6.8 Hz).

(±)-2-Formyl-3-hydroxy-5-methoxy-4-(((2*R,3*R**)-3-methyl-3-(4-methyl-2-oxopent-3-en-1-yl)oxiran-2-yl)methyl)benzyl palmitate [(±)-21a].** m-CPBA (0.1 M solution in DCM, 3.7 mL,

https://dx.doi.org/10.1021/acs.joc.0c02681
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0.37 mmol) was added to a solution of **1a** (60.9 mg, 0.107 mmol) in DCM (10.7 mL) with a syringe pump over 20 min at 23 °C (water bath). After 15 min at 23 °C, the reaction mixture was quenched by the addition of saturated aqueous $Na_2S_2O_3$ (5 mL). The resulting mixture was extracted with EtOAc (2 × 10 mL). The combined organic layer was washed with saturated aqueous $NaHCO_3$, brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 5/1 v/v) to give epoxide (±)-**21a** (44.6 mg, 0.0760 mmol, 71%). (±)-**21a**: Colorless oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.44 (1H, s, OH), 10.12 (1H, br s), 6.55 (1H, s), 6.05 (1H, br s), 5.33 (2H, s), 3.93 (3H, s), 3.07−3.01 (2H, m), 2.91−2.84 (1H, m), 2.68 (1H, d, *J* = 15 Hz), 2.43 (1H, t, *J* = 15 Hz), 2.34 (2H, t, *J* = 7.6 Hz), 2.11 (3H, d, *J* = 1.2 Hz), 1.80 (3H, d, *J* = 1.2 Hz), 1.65−1.58 (2H, m), 1.45 (3H, s), 1.32−1.22 (24H, m), 0.88 (3H, t, *J* = 7.0 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 197.8, 193.2, 173.3, 164.1, 163.5, 156.3, 139.7, 123.8, 114.0, 113.0, 105.6, 62.9, 62.1, 59.2, 56.1, 53.8, 34.3, 32.0, 29.81, 29.80, 29.79, 29.77, 29.75, 29.70, 29.56, 29.48, 29.34, 29.24, 27.8, 25.0, 22.8, 21.9, 21.0, 17.5, 14.2; FT-IR (film on ZeSe) 2924, 2855, 1744, 1682, 1628, 1458, 1381, 1250, 1134 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{35}H_{55}O_7$ [M + H]$^+$ 587.3942, found 587.3926.

(±)-2-Formyl-3-hydroxy-5-methoxy-4-(((2*R,3*R**)-3-methyl-3-(4-methyl-2-oxopent-3-en-1-yl)oxiran-2-yl)methyl)benzyl Stearate [(±)-21b].** *m*-CPBA (0.1 M solution in DCM, 5.0 mL, 0.50 mmol) was added to a solution of **1b** (101 mg, 0.168 mmol) in DCM (19 mL) through a dropping funnel over 10 min. After 2 h at room temperature, the reaction mixture was quenched by the addition of saturated aqueous $Na_2S_2O_3$ (15 mL) and extracted with EtOAc (3 × 15 mL). The combined organic layer was washed with saturated aqueous $NaHCO_3$, water, brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 7/1 v/v) to give epoxide (±)-**21b** (76.9 mg, 0.125 mmol, 74%). (±)-**21b**: reddish oil; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.44 (1H, s, OH), 10.11 (1H, s), 6.55 (1H, s), 6.05 (1H, sept, *J* = 1.0 Hz), 5.33 (2H, s), 3.92 (3H, s), 3.07−3.00 (2H, m), 2.91−2.84 (1H, m), 2.68 (1H, d, *J* = 15 Hz), 2.43 (1H, t, *J* = 15 Hz), 2.34 (2H, t, *J* = 7.5 Hz), 2.11 (3H, d, *J* = 1.0 Hz), 1.80 (3H, d, *J* = 1.0 Hz), 1.65−1.58 (2H, m), 1.44 (3H, s, H1), 1.32−1.23 (28H, m), 0.88 (3H, t, *J* = 7.0 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 197.8, 193.2, 173.3, 164.0, 163.4, 156.4, 139.6, 123.8, 114.0, 112.9, 105.6, 62.9, 62.1, 59.2, 56.1, 53.8, 34.3, 32.0, 29.81, 29.78, 29.77, 29.75, 29.70, 29.55, 29.48, 29.34, 29.22, 27.8, 25.0, 22.8, 21.8, 21.0, 17.4, 14.3; FT-IR (film on ZeSe) 2924, 2853, 1740, 1688, 1622, 1456, 1385, 1254, 1136 cm^{-1}; HRMS (APCI-pos): *m/z* calcd for $C_{37}H_{59}O_7$ [M + H]$^+$ 615.4255, found 615.4248.

Cyclization of Epoxyhericenone C [(±)-21a] with (+)-CSA (Table 3, Entry 2). (+)-CSA (27.0 mg, 0.116 mmol) was added to a solution of (±)-**21a** (27.3 mg, 0.0465 mmol) in DCM (4.7 mL) at 0 °C. After 1.5 h at 0 °C, the reaction mixture was quenched by the addition of phosphate buffer (pH 7, 3 mL) and extracted with EtOAc (2 × 5 mL). The combined organic layer was washed brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 10/1 → 3/1 v/v) to give furan **24a** (14.2 mg, 0.0250 mmol, 54%), alcohol (±)-**23a** (9.2 mg, 0.016 mol, 34%), and alcohol (±)-**22a** containing impurities (0.9 mg, <0.0015 mmol, <3%).

Cyclization of Epoxyhericenone C [(±)-21a] under Thermal Conditions (Table 3, Entry 4). A solution of (±)-**21a** (10.5 mg, 0.0179 mmol) in DCE (1 mL) was stirred for 120 h at 70 °C (small amounts of DCE were continuously added to recover loss by volatilization). The reaction mixture was directly subjected to flash chromatography (*n*-hexane/EtOAc, 10/1 → 5/1 v/v) to give alcohol (±)-**22a** (7.8 mg, 0.013 mmol, 74%).

Cyclization of Epoxyhericenone D [(±)-21b] with Benzoic Acid (Table 3, Entry 5). Benzoic acid (9.5 mg, 0.078 mmol) was added to a solution of **1b** (7.3 mg, 0.012 mmol) in DCE (1.5 mL) at room temperature. After 2 h, benzoic acid (2.9 mg, 0.024 mmol) was additionally added. After 1 h, benzoic acid (4.9 mg, 0.040 mmol) was again added. After 1 h, the reaction mixture was gradually warmed to 70 °C and stirred for 66 h (small amounts of DCE were continuously added at daily intervals to recover loss by volatilization). The reaction mixture was washed with saturated aqueous $NaHCO_3$ (2 × 5 mL), water, brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (*n*-hexane/EtOAc, 10/1 → 5/1 v/v) to give alcohol (±)-**22b** (3.6 mg, 0.0059 mmol, 49%) and furan **24b** (1.6 mg, 2.6 μmol, 22%).

Cyclization of Epoxyhericenone D [(±)-21b] with DBU (Table 3, Entry 7). DBU (18 μL, 0.12 mmol) was added to a solution of **1b** (18.7 mg, 0.0304 mmol) in DCE (1 mL) at 0 °C. The solution was stirred for 2 h at 0 °C and directly subjected to flash chromatography (*n*-hexane/EtOAc, 5/1 → 2/1 v/v) to give alcohol (±)-**22b** (14.7 mg, 0.0239 mmol, 79%) and an unstable byproduct (2.0 mg). The byproduct dissolved in $CDCl_3$ showed the signals of (±)-**23b** (0.8 mg, 1.3 μmol, 4%) and **24b** (1.2 mg, 2.0 μmol, 7%).

(±)-((*S)-7-Formyl-2-((*R**)-2-hydroxy-6-methyl-4-oxohept-5-en-2-yl)-4-methoxy-2,3-dihydrobenzofuran-6-yl)methyl Palmitate [(±)-22a].** Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.24 (1H, br s), 6.51 (1H, br s), 6.06 (1H, m), 5.48 (2H, br s), 4.93 (1H, dd, *J* = 9.6, 7.5 Hz), 4.68 (1H, s, OH), 3.89 (3H, s), 3.18 (1H, dd, *J* = 16, 7.2 Hz), 3.12 (1H, dd, *J* = 16, 9.6 Hz), 2.87 (1H, d, *J* = 17 Hz), 2.65 (1H, d, *J* = 17 Hz), 2.39 (2H, t, *J* = 7.5 Hz), 2.18 (3H, br s), 1.91 (3H, br s), 1.67 (2H, quint, *J* = 7.4 Hz), 1.33−1.23 (24H, m), 1.18 (3H, s), 0.88 (3H, t, *J* = 7.0 Hz); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 202.0, 187.8, 173.4, 166.0, 160.5, 159.1, 140.4, 124.5, 114.1, 111.5, 103.7, 89.3, 73.5, 64.4, 55.8, 49.3, 34.5, 32.1, 29.83, 29.79, 29.74, 29.61, 29.49, 29.44, 29.35, 28.1, 27.1, 25.2, 22.8, 21.7, 21.3, 14.3; FT-IR (film on ZeSe) 3460, 2924, 2855, 1740, 1678, 1612, 1497, 1462, 1423, 1404, 1377, 1319, 1288, 1258, 1207, 1161, 1126, 1045 cm^{-1}; HRMS (APCI-pos): *m/z* calcd for $C_{35}H_{55}O_7$ [M + H]$^+$ 587.3942, found 587.3932.

(±)-((2*S,3*R**)-8-Formyl-3-hydroxy-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Palmitate [(±)-23a].** Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.46 (1H, s), 6.57 (1H, br s), 6.18 (1H, sept, *J* = 1.2 Hz), 5.50 (2H, br s), 3.99 (1H, br ddd, *J* = 8.8, 5.6, 4.8 Hz), 3.88 (3H, s), 3.32 (1H, d, *J* = 4.8 Hz, OH), 2.96 (1H, dd, *J* = 17.2, 5.6 Hz), 2.91 (1H, d, *J* = 14.0 Hz), 2.86 (1H, d, *J* = 14.2 Hz), 2.53 (1H, dd, *J* = 17.2, 8.8 Hz), 2.41 (2H, t, *J* = 7.6 Hz), 2.16 (3H, d, *J* = 1.2 Hz), 1.91 (3H, d, *J* = 1.2 Hz), 1.68 (2H, quint, *J* = 7.4 Hz), 1.33 (3H, s), 1.36−1.23 (24H, m), 0.88 (3H, t, *J* = 6.8 Hz); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 199.4, 190.1, 173.4, 162.2, 159.2, 157.4, 140.3, 125.0, 115.7, 108.8, 101.5, 80.0, 67.3, 64.5, 55.8, 52.6, 34.6, 32.1, 29.83, 29.81, 29.79, 29.75, 29.61, 29.50, 29.46, 29.38, 28.2, 25.4, 25.2, 22.8, 21.3, 18.4, 14.3; FT-IR (film on ZeSe) 3454, 2925, 2853, 1741, 1671, 1605, 1575, 1466, 1423, 1402, 1378, 1302, 1213, 1155, 1119 cm^{-1}; HRMS (APCI+): *m/z* calcd for $C_{35}H_{55}O_7$ [M + H]$^+$ 587.3942, found 587.3927.

2-Formyl-3-hydroxy-5-methoxy-4-((3-methyl-5-(2-methylprop-1-en-1-yl)furan-2-yl)methyl)benzyl Palmitate (24a). Pale yellow solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 12.43 (1H, s, OH), 10.09 (1H, s, H8), 6.52 (1H, s, H1), 5.89 (2H, s, H4′, H6′), 5.31 (2H, s, H7), 3.92 (3H, s, OMe), 3.91 (2H, s, H1′), 2.33 (2H, t, *J* = 7.5 Hz, H2″), 2.02 (3H, s, H9′), 1.90 (3H, s, H10′), 1.82 (3H, s, H8′), 1.61 (2H, quint, *J* = 7.4 Hz, H3″), 1.33−1.22 (24H, m, H4″-H15″), 0.88 (3H, t, *J* = 7.0 Hz, H16″); ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 193.2 (C8), 173.4 (C1″), 163.9 (C5), 163.3 (C3), 151.1 (C5′), 146.8 (C2′), 139.2 (C1), 133.2 (7′), 116.0 (C3′), 115.0 (C4), 114.5 (C6′), 113.0 (C2), 110.8 (C4′), 105.6 (C6), 63.0 (C7), 56.1 (OMe), 34.4 (C2″), 32.1, 29.84, 29.83, 29.82, 29.80, 29.78, 29.73, 29.58, 29.51, 29.36, 29.24, 27.1 (C8′), 25.0, 22.8, 19.9 (C10′), 19.2 (C1′), 14.3 (C16″), 10.0 (C9′); FT-IR (film on ZeSe) 2924, 2853, 1742, 1634, 1580, 1462, 1416, 1300, 1254, 1221, 1140 cm^{-1}; HRMS (ESI-pos): *m/z* calcd for $C_{35}H_{53}O_6$ [M + H]$^+$ 569.3837, found 569.3820.

(±)-((*S)-7-Formyl-2-((*R**)-2-hydroxy-6-methyl-4-oxohept-5-en-2-yl)-4-methoxy-2,3-dihydrobenzofuran-6-yl)methyl Stearate [(±)-22b].** Colorless solid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 10.23 (1H, br s), 6.51 (1H, br s), 6.06 (1H, sept, *J* = 1.1 Hz), 5.48 (2H, br s), 4.94 (1H, dd, *J* = 9.6, 7.5 Hz), 4.74 (1H, s, OH), 3.89 (3H, s), 3.18 (1H, dd, *J* = 16, 7.2 Hz), 3.12 (1H, dd, *J* = 16, 9.6 Hz), 2.88 (1H, d, *J* = 17 Hz), 2.65 (1H, d, *J* = 17 Hz), 2.39 (2H, t, *J* = 7.5 Hz), 2.18 (3H, d, *J* = 1.1 Hz), 1.91 (3H, d, *J* = 1.1 Hz), 1.67 (2H,

https://dx.doi.org/10.1021/acs.joc.0c02681
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quint, J = 7.4 Hz), 1.34−1.23 (28H, m), 1.17 (3H, s), 0.87 (3H, t, J = 6.8 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 202.0, 187.8, 173.4, 166.0, 160.5, 159.1, 140.4, 124.5, 114.1, 111.5, 103.7, 89.3, 73.5, 64.4, 55.8, 49.3, 34.5, 32.1, 29.85, 29.83, 29.81, 29.76, 29.62, 29.51, 29.46, 29.37, 28.1, 27.1, 25.2, 22.8, 21.7, 21.3, 14.3; FT-IR (film on ZeSe) 3460, 2916, 2851, 1740, 1674, 1612, 1466, 1381, 1292, 1258, 1211, 1177, 1126 cm^{-1}; HRMS (APCI+): m/z calcd for $C_{37}H_{59}O_7$ $[M + H]^+$ 615.4255, found 615.4257.

(±)-((2*S,3*R**)-3-Acetoxy-8-formyl-5-methoxy-2-methyl-2-(4-methyl-2-oxopent-3-en-1-yl)chroman-7-yl)methyl Palmitate [(±)-25].** Pyridine (19 μL, 0.23 mmol) and Ac_2O (11 μL, 0.12 mmol) were added to a solution of alcohol (±)-**23** (7.2 mg, 0.012 mmol) and DMAP (1.4 mg, 0.011 mmol) in DCM (1 mL). After 1 h at room temperature, the reaction mixture was directly subjected to flash chromatography on silica gel (n-hexane/EtOAc, 10/1 → 5/1 v/v) to give acetate (±)-**25** (7.4 mg, 0.012 mmol, 100%). (±)-**25**: Yellow oil; ^1H NMR (400 MHz, $CDCl_3$) δ 10.41 (1H, s), 6.58 (1H, s), 6.01 (1H, qq, J = 1.2, 0.8 Hz), 5.52 (2H, s), 5.28 (1H, t, J = 5.1 Hz), 3.88 (3H, s), 2.96 (1H, dd, J = 18, 5.1 Hz), 2.80 (1H, d, J = 14 Hz), 2.71 (1H, dd, J = 18, 5.1 Hz), 2.61 (1H, d, J = 14 Hz), 2.42 (2H, t, J = 7.6 Hz), 2.14 (3H, d, J = 0.8 Hz), 2.08 (3H, s), 1.86 (3H, s, J = 1.2 Hz), 1.69 (2H, quint, J = 7.6 Hz), 1.45 (3H, s), 1.36−1.23 (24H, m), 0.87 (3H, t, J = 6.8 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 196.7 (C2′), 190.2 (C10), 173.4 (C1″), 170.4 (CH_3CO), 162.2 (C5), 157.6 (C4′), 157.2 (C8a), 140.5 (C7), 124.8 (C3′), 115.8 (C8), 106.9 (C4a), 101.4 (C6), 77.7 (C2), 68.6 (C3), 64.6 (C9), 55.8 (OMe), 50.5 (C1′), 34.6 (C2″), 32.1, 29.83, 29.82, 29.80, 29.76, 29.62, 29.50, 29.47, 29.39, 28.0 (C5′), 25.3 (C3″), 23.3 (C4), 22.8, 21.3 (CH_3CO), 21.1 (C6′), 20.4 (C11), 14.3 (C16″); FT-IR (film on ZeSe) 2924, 2855, 1744, 1678, 1609, 1574, 1462, 1377, 1300, 1234, 1157, 1119, 1045 cm^{-1}; HRMS (APCI+): m/z calcd for $C_{37}H_{57}O_8$ $[M + H]^+$ 629.4048, found 629.4043.

(*E*)-4-(3,7-Dimethylocta-2,6-dien-1-yl)-2-(hydroxymethyl)-5-methoxy-3-((2-(trimethylsilyl)ethoxy)methoxy)benzyl Linoleate (28) and (*E*)-3-(3,7-Dimethylocta-2,6-dien-1-yl)-6-(hydroxymethyl)-4-methoxy-2-((2-(trimethylsilyl)ethoxy)methoxy)benzyl Linoleate (29). NaH (16.5 mg, 0.688 mmol), SEMCl (108 μL, 0.618 mmol), and n-Bu_4NBr (10.4 mg, 0.0323 mmol) were added to a solution of hericene D (**3d**) (120 mg, 0.206 mmol) in THF (10 mL) at 0 °C. Two drops of water (5 μL × 2) were added at the intervals of 10 min, and the mixture was stirred for 0.5 h at 0 °C. Additional NaH (4.9 mg, 0.21 mmol), SEMCl (36 μL, 0.21 mmol), n-Bu_4NBr (6.6 mg, 0.021 mmol), and water (5 μL) were added, and the stirring was continued for 0.2 h. The reaction mixture was diluted with n-hexane (20 mL) and quenched by the addition of saturated aqueous NH_4Cl (5 mL). The resulting mixture was extracted with n-hexane (3 × 10 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated after the addition of Et_3N (2 mL). The residue was roughly purified by flash chromatography (n-hexane/EtOAc/Et_3N, 20/1/0.4 v/v) to give a corresponding SEM ether (110 mg, 0.154 mmol, 75%) that contained a small amount of impurities. $NaBH_4$ (15.2 mg, 0.402 mmol) was added to a solution of SEM ether (110 mg, 0.154 mmol) in THF (8 mL). After 1 h at room temperature, the reaction mixture was warmed to 35 °C with an aluminum heating bath and stirred for 1 h. Additional $NaBH_4$ (12.3 mg + 11.3 mg, 0.325 mmol +0.299 mmol) was added at the intervals of 1 h at 35 °C, and the mixture was stirred for 2.7 h in total. The reaction mixture was quenched by the addition of saturated aqueous NH_4Cl at 0 °C and extracted with EtOAc (3 × 10 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (n-hexane/EtOAc, 15/1 → 10/1 v/v) to give alcohols **28** (62.2 mg, 0.0872 mmol, 57%) and **29** (38.4 mg, 0.0538 mmol, 35%). **28**: colorless oil; ^1H NMR (400 MHz, $CDCl_3$) δ 6.74 (1H, s), 5.42−5.28 (4H, m), 5.24 (2H, s), 5.11 (1H, br t, J = 6.8 Hz), 5.04 (1H, m), 5.01 (2H, s), 4.63 (2H, br s), 3.87 (2H, m) 3.83 (3H, s), 3.36 (1H, br s, OH), 3.32 (2H, d, J = 6.8 Hz), 2.76 (2H, t, J = 6.6 Hz), 2.34 (2H, t, J = 7.4 Hz), 2.08−1.93 (8H, m), 1.75 (3H, s), 1.64 (3H, s), 1.61 (2H, m), 1.57 (3H, s), 1.38−1.24 (14H, m), 1.04 (2H, m), 0.88 (3H, t, J = 6.8 Hz), 0.03 (9H, s); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 173.7, 158.2, 156.5, 135.4, 134.2, 131.5, 130.3, 130.2, 128.2, 128.0, 126.7, 124.3, 123.9, 122.6, 108.9, 98.7, 68.0, 64.2, 56.1, 55.8, 39.8, 34.5, 31.7, 29.73, 29.48, 29.31, 29.25, 27.3, 26.7, 25.81, 25.75, 25.0, 23.8, 22.7, 18.1, 17.8, 16.4, 14.2, −1.4; FT-IR (film on ZeSe) 3507, 3007, 2928, 2855, 1738, 1605, 1582, 1464, 1454, 1402, 1379, 1296, 1157, 1113 cm^{-1}; HRMS (APCI−): m/z calcd for $C_{43}H_{72}O_6ClSi$ $[M + Cl]^-$ 747.4792, found 747.4821. **29**: colorless oil; ^1H NMR (400 MHz, $CDCl_3$) δ 6.83 (1H, s), 5.42−5.28 (4H, m), 5.27 (2H, s), 5.14 (1H, br t, J = 6.4 Hz), 5.05 (1H, m), 4.98 (2H, s), 4.70 (2H, br s), 3.85 (2H, m) 3.85 (3H, s), 3.36 (2H, d, J = 6.4 Hz), 2.76 (2H, t, J = 6.6 Hz), 2.28 (2H, t, J = 7.6 Hz), 2.17 (1H, br s, OH), 2.07−1.93 (8H, m), 1.75 (3H, br s), 1.64 (3H, s), 1.61 (2H, m), 1.58 (3H, s), 1.37−1.25 (14H, m), 1.00 (2H, m), 0.88 (3H, t, J = 6.8 Hz), 0.03 (9H, s); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 174.0, 159.3, 156.4, 140.1, 135.3, 131.4, 130.4, 130.2, 128.2, 128.0, 124.4, 123.7, 122.8, 119.4, 107.5, 99.2, 67.8, 63.3, 59.2, 55.8, 39.8, 34.6, 31.7, 29.75, 29.49, 29.33, 29.27, 27.3, 26.8, 25.83, 25.76, 25.1, 23.7, 22.7, 18.3, 17.8, 16.4, 14.2, −1.3; FT-IR (film on ZeSe) 3451, 3009, 2955, 2926, 2857, 1736, 1605, 1582, 1466, 1379, 1250, 1190, 1159, 1107 cm^{-1}; HRMS (APCI−): m/z calcd for $C_{43}H_{72}O_6ClSi$ $[M + Cl]^-$ 747.4792, found 747.4827.

(*E*)-3-(3,7-Dimethylocta-2,6-dien-1-yl)-6-formyl-2-hydroxy-4-methoxybenzyl Linoleate (4d). $PhI(OAc)_2$ (40.2 mg, 0.125 mmol) and AZADOL (0.6 mg, 0.004 mmol) were added to a solution of alcohol **29** (29.7 mg, 0.0416 mmol) in DCM (2.1 mL). After 3 h at room temperature, the reaction mixture was quenched by the addition of saturated aqueous $Na_2S_2O_3$ and saturated aqueous $NaHCO_3$ and extracted with n-hexane (2 × 10 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (n-hexane/EtOAc, 10/1 v/v) to give the corresponding aldehyde (30.3 mg, quant) as a colorless oil. ^1H NMR (400 MHz, $CDCl_3$) δ 10.24 (1H, s), 7.27 (1H, s), 5.50 (2H, s), 5.41−5.28 (4H, m), 5.12 (1H, br t, J = 6.8 Hz), 5.04 (1H, m), 5.00 (2H, s), 3.89 (3H, s), 3.86 (2H, m), 3.43 (2H, d, J = 6.8 Hz), 2.76 (2H, t, J = 6.6 Hz), 2.28 (2H, t, J = 7.8 Hz), 2.08−1.94 (8H, m), 1.76 (3H, d, J = 0.8 Hz), 1.63 (3H, d, J = 0.8 Hz), 1.61 (2H, m), 1.56 (3H, br s), 1.37−1.24 (14H, m), 1.00 (2H, m), 0.88 (3H, t, J = 6.8 Hz), 0.03 (9H, s); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 190.7, 173.5, 159.2, 156.4, 136.3, 134.2, 131.5, 131.3, 130.4, 130.2, 128.1, 128.0, 124.4, 124.3, 121.5, 107.0, 99.4, 68.1, 57.1, 56.0, 39.8, 34.4, 31.7, 29.74, 29.49, 29.30, 29.26, 27.3, 26.7, 25.82, 25.75, 25.0, 24.3, 22.7, 18.3, 17.8, 16.4, 14.2, −1.3; FT-IR (film on ZeSe) 3007, 2926, 2857, 1738, 1694, 1593, 1578, 1464, 1400, 1377, 1292, 1250, 1159, 1111 cm^{-1}; HRMS (APCI-neg): m/z calcd for $C_{43}H_{70}O_6ClSi$ $[M + Cl]^-$ 745.4636, found 745.4671. A solution of $MgBr_2$ (73.5 mg, 0.399 mmol) in Et_2O (2.5 mL) and $MeNO_2$ (0.25 mL) was added to a solution of aldehyde (30.3 mg) in Et_2O (0.5 mL) at 0 °C. The mixture was warmed to room temperature and stirred for 4 h. The reaction mixture was diluted with n-hexane (5 mL) and quenched by the addition of water (5 mL). After separation of the organic layer, the aqueous phase was extracted with EtOAc (2 × 5 mL). The combined organic layer was washed with brine, dried over anhydrous $MgSO_4$, filtered, and concentrated. The residue was purified by flash chromatography (n-hexane/EtOAc, 10/1 → 2/1 v/v) to give the regioisomer of hericene D (**4d**) (11.3 mg, 0.0195 mmol, 47% for two steps). **4d**: colorless oil; ^1H NMR (400 MHz, $CDCl_3$) δ 10.20 (1H, s), 8.40 (1H, s), 7.01 (1H, s), 5.50 (2H, s), 5.42−5.28 (4H, m), 5.19 (1H, br t, J = 7.6 Hz), 5.05 (1H, m), 3.89 (3H, s), 3.46 (2H, d, J = 7.2 Hz), 2.76 (2H, t, J = 6.0 Hz), 2.35 (2H, t, J = 7.6 Hz), 2.07−1.95 (8H, m), 1.79 (3H, s), 1.64 (3H, s), 1.59 (2H, m), 1.57 (3H, s), 1.36−1.21 (14H, m), 0.88 (3H, t, J = 6.8 Hz); $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$) δ 191.8, 177.0, 159.1, 155.7, 136.9, 133.9, 131.5, 130.4, 130.2, 128.2, 128.0, 125.7, 124.4, 121.0, 116.3, 106.7, 57.7, 55.9, 39.9, 34.3, 31.7, 29.69, 29.49, 29.22, 29.19, 29.10, 27.34, 27.31, 26.76, 25.83, 25.76, 24.9, 23.4, 22.7, 17.8, 16.3, 14.2; FT-IR (film on ZeSe) 3244, 3009, 2928, 2855, 1732, 1701, 1601, 1578, 1504, 1462, 1431, 1400, 1381, 1319, 1273, 1231, 1172, 1119 cm^{-1}; HRMS (APCI-pos): m/z calcd for $C_{37}H_{57}O_5$ $[M + H]^+$ 581.4201, found 581.4205.

https://dx.doi.org/10.1021/acs.joc.0c02681
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Materials for Neuroprotective Assay. Neuro2a cells were obtained from the Health Science Research Resources Bank, Japan, and maintained in Dulbecco's modified Eagle's medium (D-MEM) (Sigma, USA) supplemented with 10% fetal bovine serum (FBS) at 37 °C in a humidified atmosphere containing 5% CO_2 in air, unless particularly noted. Tunicamycin (TM) and thapsigargin (TG) were obtained from Nacalai Tesque and SIGMA, respectively, and were dissolved in D-MEM without FBS at concentrations of 5.0 μg/mL for TM and 100 nM for TG. The test samples were first completely dissolved in DMSO (100 mM) and then diluted with D-MEM without FBS to 3−100 μM. Visually, no precipitation was detected at this point. For cytological staining, commercially available Calcein-AM, propidium iodide, and Hoechst 33342 were used to stain living cells, dead cells, and nuclei, respectively.

Cell Protection Assay. Neuro2a cells were cultured in D-MEM with 10% (v/v) FBS (70 μL) for 22−24 h in 96-well plates, starting at a cell density of 2000, 3000, or 5000 cells/well. The medium was removed, and immediately, the cells were pretreated with varying concentrations (3−100 μM) of test compounds in D-MEM without FBS (70 μL) for 2 h. Next, the ER stressors were added to the medium (final concentration: 0.5 μg/mL for TM and 10 nM for TG) and the cells were incubated for additional 22 h. After this treatment, cell viability was measured by using the 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2*H*-tetrazolium bromide (MTT) assay as described previously.[67] Briefly, the medium was replaced by 0.25 mg/mL of MTT in D-MEM without FBS (70 μL), and the cells were incubated for 2 h. Incubation was terminated by the addition of 15% (w/v) sodium dodecyl sulfate and 50% (v/v) *N*,*N*-dimethylformamide in water (70 μL). The absorbance of the reaction mixture at 570 nm was measured by a microplate reader.

Statistical Analysis. Data collected were analyzed statistically using Tukey−Kramer multiple comparisons tests to determine significant differences in the data among the groups. p values less than 0.01 or 0.05 were considered significant. The values are expressed as means ± SD.

Cell Staining. Neuro2a cells were cultured in D-MEM with 10% (v/v) FBS (420 μL) for 22 h in 24-well plates, starting at a cell density of 5000 cells/well. The medium was removed, and immediately, the cells were pretreated with **4d** (30 μM) in D-MEM without FBS (420 μL) for 2 h. Next, the ER stressors were added to the medium (final concentration: 0.5 μg/mL for TM and 10 nM for TG) and the cells were incubated for additional 22 h. After the treatment, the medium was replaced by Calcein-AM (1 μg/mL), propidium iodide (1 μg/mL), or Hoechst 33342 (1 μg/mL) in D-MEM without FBS (300 μL), and the cells were incubated for 0.5 h. The stained cells were photographed by a fluorescence microscope (ZOE Fluorescent Cell Imager, Bio-Rad).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c02681.

> TLC analyses for cyclization of (±)-**21a**/**b**; comparisons of experimental and predicted ^{13}C NMR data for (±)-**22** and (±)-**23**; comparisons of NMR data of natural products and synthetic compounds; a conformational analysis of acetate **25**; neuroprotective effects of **1b**, **1c**, and (±)-**2b**; and NMR spectra of all synthetic compounds (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Number JP17K07779. We thank Prof. Araki Masuyama for his generous support on this project, and Mr. Hisashi Fujita, Mr. Hiroki Matsui, Mr. Tatsuya Takumi, and Mr. Masanori Kawaguchi for preparation of intermediates. We also thank Ms. Sayaka Kado, Center for Analytical Instrumentation, Chiba University, for measurement of mass spectra.

■ DEDICATION

This work is dedicated to Prof. Ilhyong Ryu (Osaka Prefecture University; National Chiao Tung University) on the occasion of his 70th birthday.

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(45) The use of ethereal solvent such as THF, 1,4-dioxane, or CPME gave inferior conversions or selectivity.
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(54) Presence of the OH signal was highly dependent on the purity and concentration of the sample. In many cases taking NMR spectra, the OH signal was not observed and the C3−H was observed as doublet of doublets.
(55) In the case of acetate (±)-**25**, NOESY correlation was observed for C3−H and C2−Me. This observation probably arises from the conformation of the acetate in which the C3−H and C2−Me occupy pseudo-equatorial orientations. A reasonable conformation deduced from the molecular mechanics calculation and observed coupling constants for C3−H and C4−H (J_{H3-H4} = 5.2 Hz × 2) is presented in Figure S4 in the Supporting Information.
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(64) Enlarged pictures in Figure 4 indicate an enhancement of the neuronal network when unstressed cells were treated with **4d**. More detailed studies are needed to elucidate the mechanism of action for **4d**.
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https://dx.doi.org/10.1021/acs.joc.0c02681
J. Org. Chem. 2021, 86, 2602−2620